# Southeast Conservation Blueprint Summary

for Texas

**Created 10/16/2026**

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**The Southeast Conservation Blueprint 2026**

Southeast Conservation Adaptation Strategy

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# About the Southeast Blueprint

The Southeast Conservation Blueprint is the primary product of the Southeast Conservation Adaptation Strategy (SECAS). It is a living, spatial plan to achieve the SECAS vision of a connected network of lands and waters across the Southeast and Caribbean. The Blueprint is regularly updated to incorporate new data, partner input, and information about on-the-ground conditions.

The Blueprint identifies priority areas based on a suite of natural and cultural resource indicators representing terrestrial, freshwater, and marine ecosystems. A connectivity analysis identifies corridors that link coastal and inland areas and span climate gradients.

For more information:

- Visit the Blueprint webpage
- Review the Blueprint 2026 Development Process
- View and download the Blueprint data and make maps on the Blueprint page of the SECAS Atlas

## We're here to help!

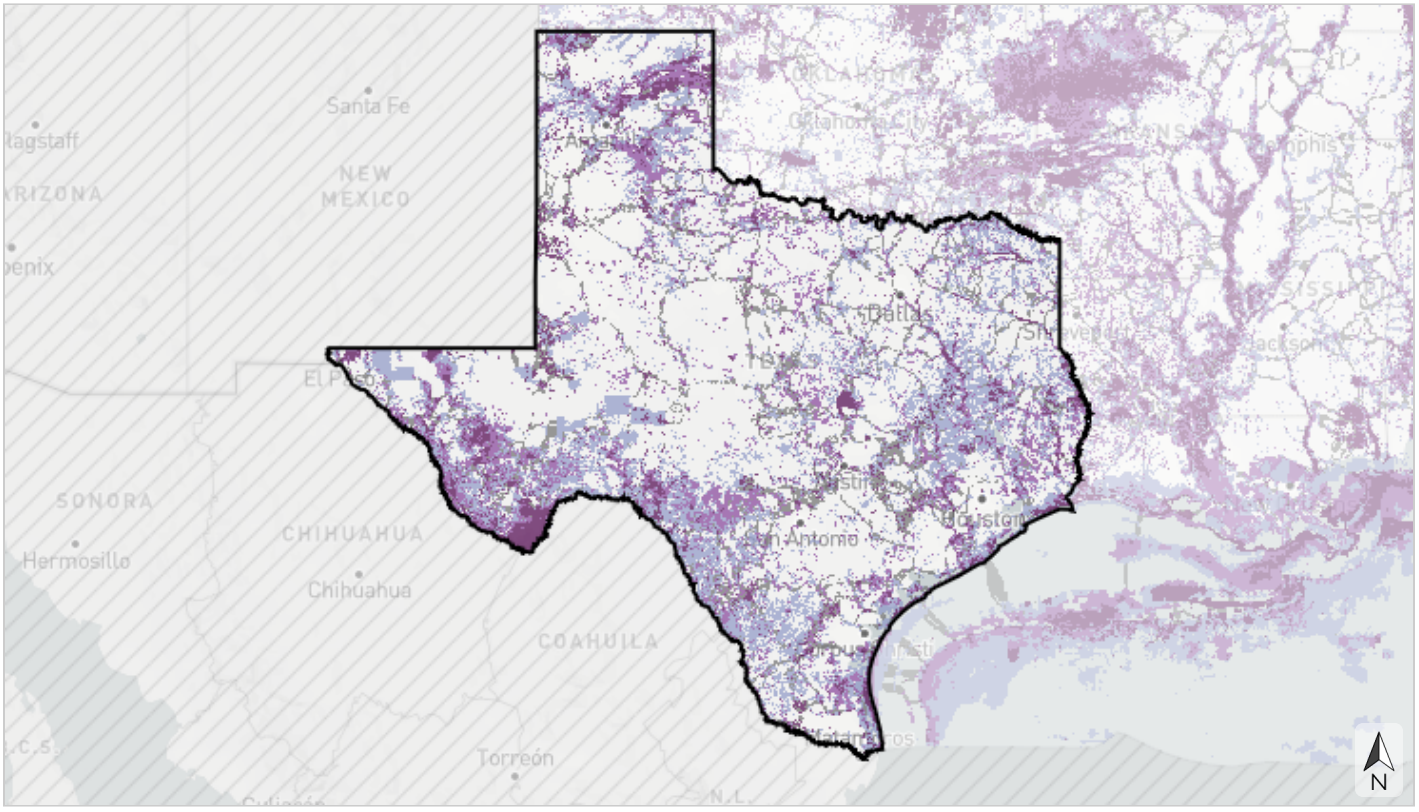
- Do you have a question about the Blueprint?
- Would you like help using the Blueprint to support a proposal or inform a decision?
- Do you have a suggestion on how to improve the Blueprint? The Blueprint and its inputs are regularly revised based on input from people like you.
- Do you have feedback on how to improve the Blueprint Explorer interface?

If you need help or have questions, contact Southeast Blueprint staff by reaching out to a member of the user support team.

We're here to support you. We really mean it. It's what we do!

This report was generated by the Southeast Conservation Blueprint Explorer, which was developed by Astute Spruce, LLC in partnership with the U.S. Fish and Wildlife Service under the Southeast Conservation Adaptation Strategy.

# Southeast Blueprint Priorities




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**Priorities for a connected network of lands and waters**

- Highest priority
- High priority
- Medium priority
- Priority connections

# Priority Categories

## For a connected network of lands and waters

In total, Blueprint priorities and priority connections cover roughly 50% of the Southeast Blueprint geography.

### Highest priority

Areas where conservation action would make the biggest impact, based on a suite of natural and cultural resource indicators. This class covers roughly 10% of the Southeast Blueprint geography.

### High priority

Areas where conservation action would make a big impact, based on a suite of natural and cultural resource indicators. This class covers roughly 15% of the Southeast Blueprint geography.

### Medium priority

Areas where conservation action would make an above-average impact, based on a suite of natural and cultural resource indicators. This class covers roughly 20% of the Southeast Blueprint geography.

### Priority connections

Connections between priority areas that cover the shortest distance possible while routing through as much Blueprint priority as possible. This class covers roughly 5% of the Southeast Blueprint geography.

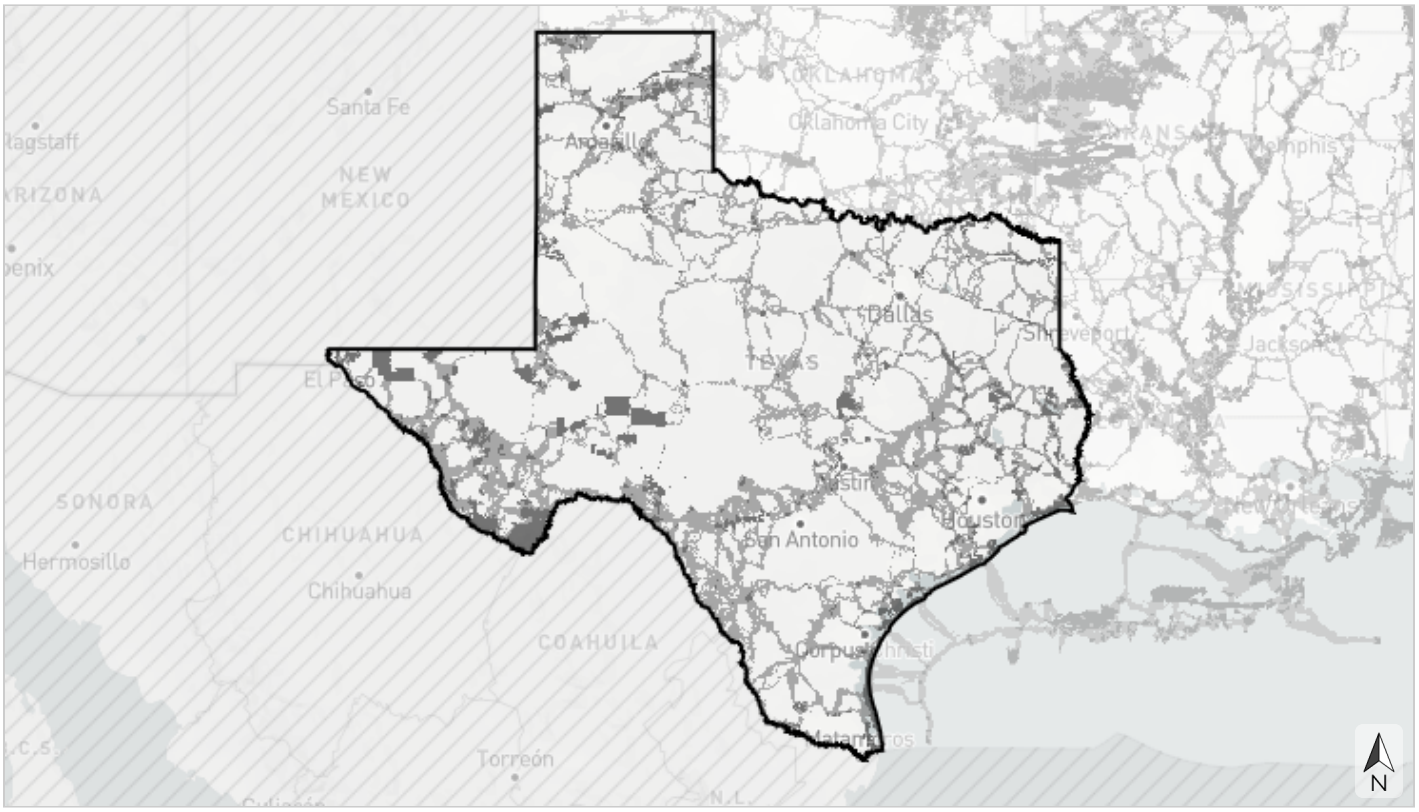
*Table 1: Extent of each Blueprint priority category within Texas.*

| Priority Category | Acres | Percent of Area |
|---|---|---|
| Highest priority | 14,176,226 | 8.2% |
| High priority | 24,378,112 | 14.2% |
| Medium priority | 35,164,584 | 20.5% |
| Priority connections | 11,874,109 | 6.9% |
| Lower priority | 86,304,394 | 50.2% |
| **Total area** | **171,897,425** | **100%** |

# Hubs and Corridors

The Blueprint uses a least-cost path connectivity analysis to identify corridors that link hubs across the shortest distance possible, while also routing through as much Blueprint priority as possible.

In the continental Southeast, hubs are large patches (~5,000+ acres) of highest priority Blueprint areas and/or protected lands.




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Hubs
Corridors

*Table 2: Extent of hubs and corridors within Texas.*

| Type | Acres | Percent of Area |
|---|---|---|
| Hubs | 10,493,813 | 6.1% |
| Corridors | 37,093,591 | 21.6% |
| Not a hub or corridor | 124,310,021 | 72.3% |
| **Total area** | **171,897,425** | **100%** |

# Indicator Summary

*Table 3: Terrestrial indicators.*

| Indicator | Present |
|---|---|
| Fire frequency | ✓ |
| Grassland & savanna extent | ✓ |
| Grassland & savanna restoration | ✓ |
| Greenways & trails | ✓ |
| Imperiled amphibians & reptiles | ✓ |
| Imperiled mammals | ✓ |
| Intact habitat cores | ✓ |
| Landscape condition | ✓ |
| Mississippi Alluvial Valley forest birds - protection | - |
| Mississippi Alluvial Valley forest birds - reforestation | - |
| Playas | ✓ |
| Potential access to parks | ✓ |
| Resilient terrestrial sites | ✓ |
| River cane restoration | ✓ |
| Urban park size | ✓ |
| West Coastal Plain & Ouachitas forested wetland birds | ✓ |
| West Coastal Plain & Ouachitas open pine birds | ✓ |
| West Gulf Coast mottled duck nesting | ✓ |

*Table 4: Freshwater indicators.*

| Indicator | Present |
|---|---|
| Floodplain inundation | ✓ |
| Gulf migratory fish connectivity | - |
| Imperiled aquatic species | ✓ |
| Lakes & reservoirs | ✓ |
| Natural landcover in floodplains | ✓ |
| Network complexity | ✓ |
| Permeable surface | ✓ |

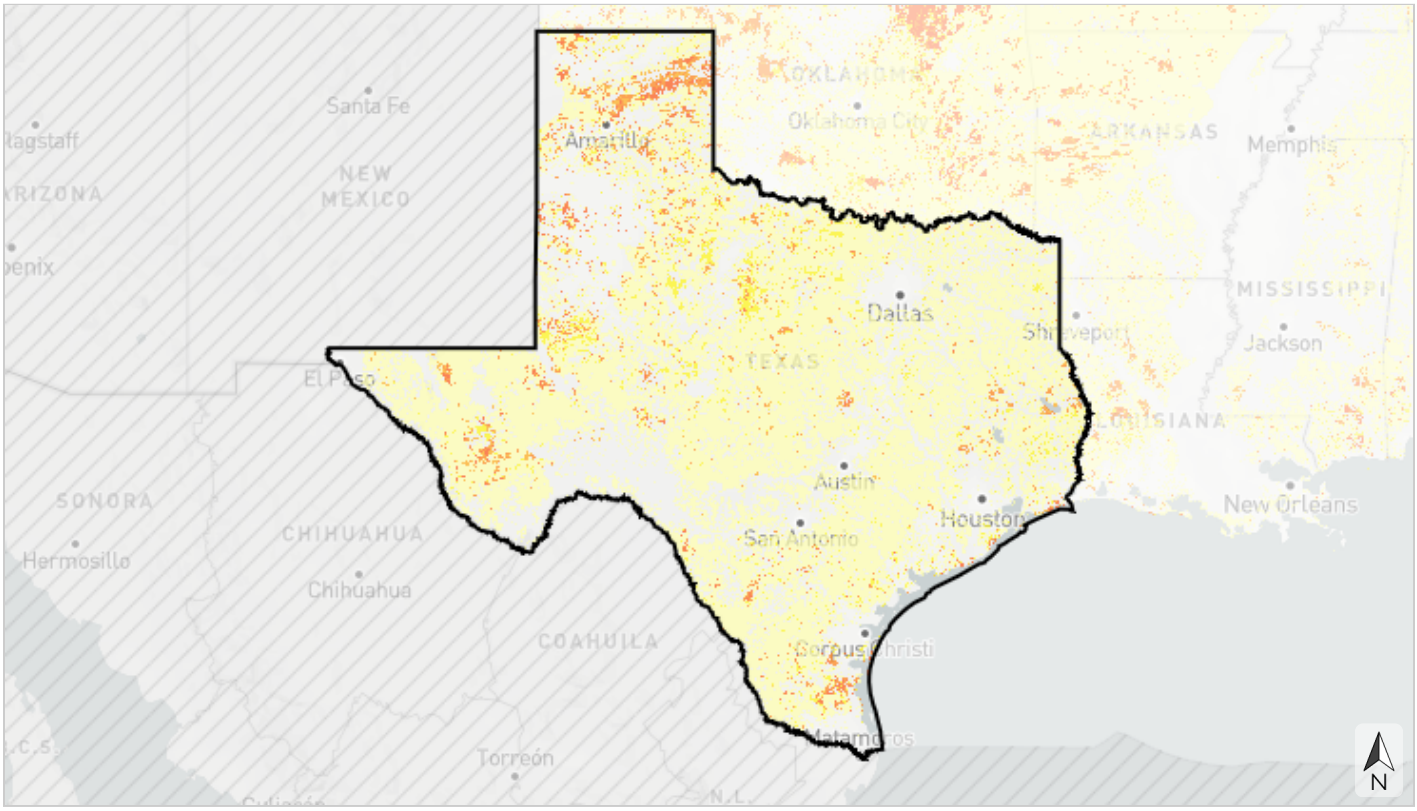
*Table 5: Coastal & marine indicators.*

| Indicator | Present |
|---|---|
| Coastal shoreline condition | ✓ |
| Estuarine coastal condition | ✓ |
| Gulf coral & hardbottom | ✓ |
| Gulf deep-sea coral richness | - |
| Gulf marine birds | ✓ |
| Gulf marine mammals | ✓ |
| Gulf sea turtles | ✓ |
| Island habitat | ✓ |
| Marine highly migratory fish | ✓ |
| Resilient coastal sites | ✓ |
| Seagrass | ✓ |
| South Atlantic maritime forest | - |
| Stable coastal wetlands | ✓ |

Terrestrial

# Fire frequency

This indicator uses remote sensing, fire treatment data, and historical landcover models to assess whether an area matches its historical fire regime now and in the recent past (1984-2024). It focuses on fire-adapted forests and grasslands that historically burned at least once every 20 years, which rely on regular, low-intensity fires to maintain habitat, support native plant growth, and reduce wildfire risk. It evaluates departure from historical conditions based on the U.S. Forest Service's fire regime condition classes. This indicator combines Landsat-based fire history metrics from the U.S. Geological Survey, fire treatment data from LANDFIRE, and historical fire regime estimates from LANDFIRE. Development of fire history metrics was supported by Southeast FireMap.




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**How well fire-adapted areas match their historic fire regime under current conditions vs. recent past (1984-2024)**

- Currently in historical fire regime, low (<33%) departure in recent past
- Currently in historical fire regime, medium (33-66%) departure in recent past
- Currently in historical fire regime, high (>66%) departure in recent past
- Not currently in historical fire regime, low (<33%) departure in recent past
- Not currently in historical fire regime, medium (33-66%) departure in recent past
- Not currently in historical fire regime, high (>66%) departure in recent past
- Urban, row crop, or not a fire-adapted area (>20-year fire regime)

*Table 6: Indicator values for fire frequency within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

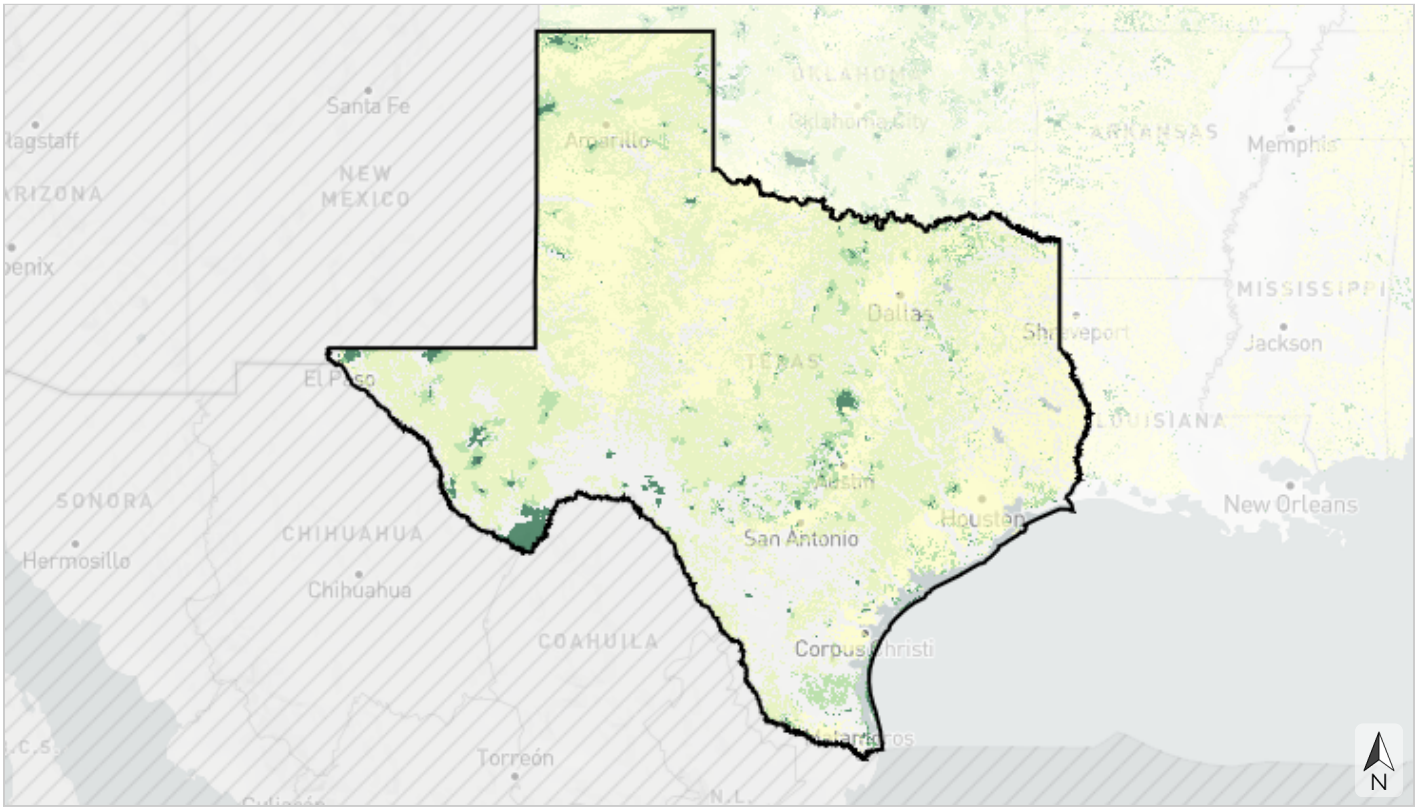
| | Indicator Values: How well fire-adapted areas match their historic fire regime under current conditions vs. recent past (1984-2024) | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Currently in historical fire regime, low (<33%) departure in recent past | 1,527,198 | 0.9% | |
| | Currently in historical fire regime, medium (33-66%) departure in recent past | 4,320,908 | 2.5% | ↑ In good condition |
| | Currently in historical fire regime, high (>66%) departure in recent past | 1,800,477 | 1.0% | ↓ Not in good condition |
| | Not currently in historical fire regime, low (<33%) departure in recent past | 281,538 | 0.2% | |
| | Not currently in historical fire regime, medium (33-66%) departure in recent past | 6,711,163 | 3.9% | |
| | Not currently in historical fire regime, high (>66%) departure in recent past | 85,853,904 | 49.9% | |
| ↓ Low | Urban, row crop, or not a fire-adapted area (>20-year fire regime) | 71,361,835 | 41.5% | |
| | *Area not evaluated for this indicator* | *40,404* | *<0.1%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Grassland & savanna extent

This indicator represents grasslands and savannas in the southeastern United States, which support important plants, reptiles, amphibians, mammals, birds, and pollinators. It considers known grassland and savanna locations, predicted locations within protected areas, and surrounding pollinator buffers. It also incorporates other potential grassland and savanna locations within natural and altered landscapes, and restoration opportunities within historical locations based on past fire intervals and historic ecosystem predictions. This indicator combines data from multiple sources, including the Southeastern Grasslands Institute, the National Land Cover Database, LANDFIRE, Oklahoma and Texas ecological systems maps, and more.




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- Known grassland/savanna
- Likely grassland/savanna >10 acres
- Likely grassland/savanna ≤10 acres
- Pollinator buffer around known or likely grassland/savanna
- Potential grassland/savanna in grassland/savanna hub
- Potential grassland/savanna outside grassland/savanna hub
- Historic grassland/savanna
- Not identified as grassland/savanna

*Table 7: Indicator values for grassland & savanna extent within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

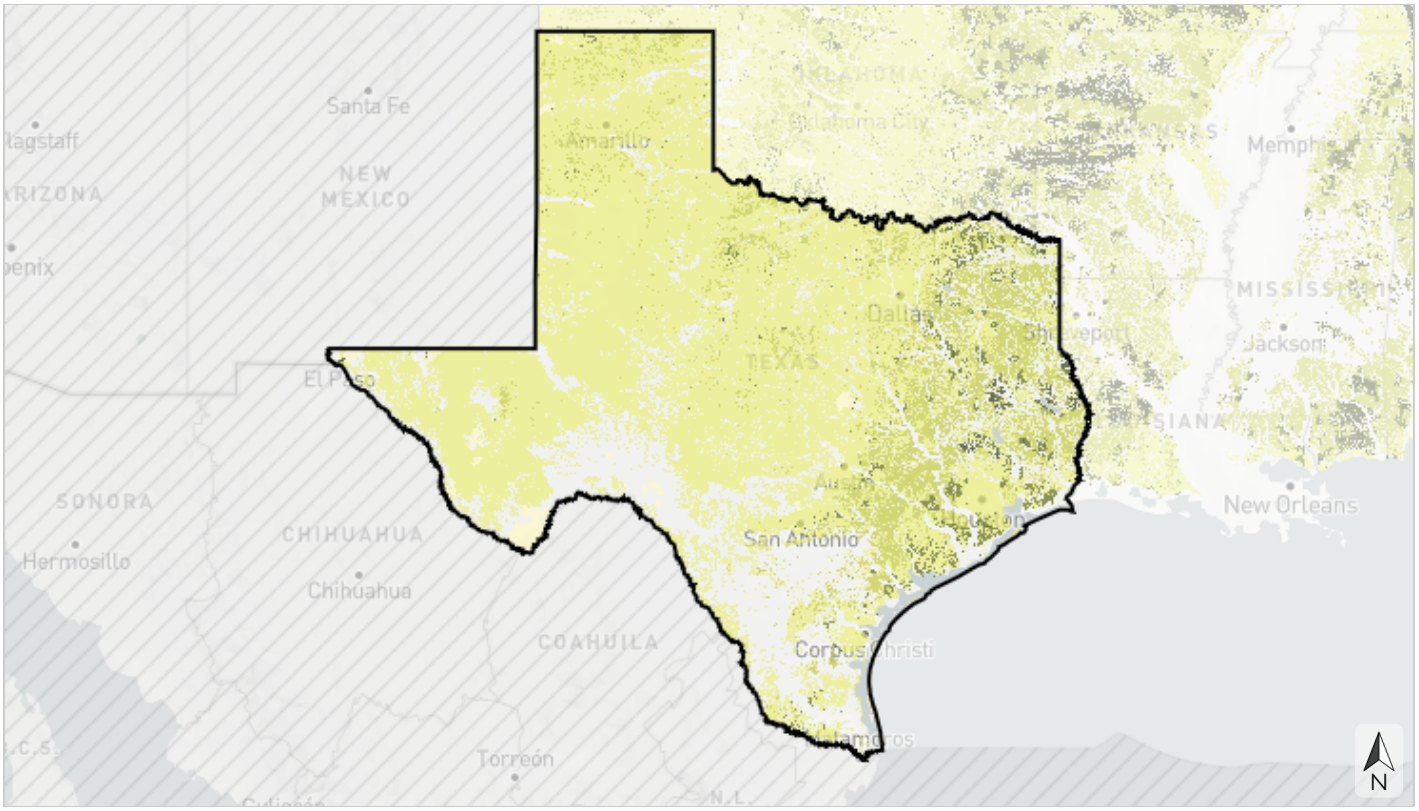
| | Indicator Values | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Known grassland/savanna | 27,985 | <0.1% | |
| | Likely grassland/savanna >10 acres | 2,336,216 | 1.4% | |
| | Likely grassland/savanna ≤10 acres | 34,573 | <0.1% | ↑ In good condition |
| | Pollinator buffer around known or likely grassland/savanna | 1,023,451 | 0.6% | ↓ Not in good condition |
| | Potential grassland/savanna in grassland/savanna hub | 8,913,746 | 5.2% | |
| | Potential grassland/savanna outside grassland/savanna hub | 58,213,894 | 33.9% | |
| | Historic grassland/savanna | 49,868,644 | 29.0% | |
| ↓ Low | Not identified as grassland/savanna | 51,460,371 | 29.9% | |
| | *Area not evaluated for this indicator* | *18,545* | *<0.1%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Grassland & savanna restoration

This indicator represents potential restoration opportunities for grasslands and savannas within their historical range in the southeastern United States. Grasslands support important plants, reptiles, amphibians, mammals, birds, and pollinators, but have significantly declined from their current extent. It considers proximity to clusters of existing grasslands and savannas, protected areas with management potential, landcover classes with good restoration potential, priority areas for restoration of longleaf and shortleaf pine, and historical grassland and savanna locations. This indicator combines data from multiple sources, including the Southeast Blueprint grassland and savanna extent indicator, the Longleaf Suitability Analysis, shortleaf pine suitability, landcover from the National Land Cover Database and LANDFIRE, and more.




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**Restoration priority**

- Highest (in a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres)
- Very high (in a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres)
- High (outside a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres)
- Medium (outside a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres)
- Low (other historic grassland/savanna)
- Very low (already known or likely grassland/savanna)
- Lowest (not identified as historic grassland/savanna)

*Table 8: Indicator values for grassland & savanna restoration within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values: Restoration priority | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Highest (in a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres) | 505,111 | 0.3% |
| | Very high (in a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres) | 3,376,368 | 2.0% |
| | High (outside a grassland/savanna hub, in a protected area with management potential, in a historic patch >10 acres) | 174,921 | 0.1% |
| | Medium (outside a grassland/savanna hub, unprotected landcover with good restoration potential, in a historic patch >10 acres) | 17,836,943 | 10.4% |
| | Low (other historic grassland/savanna) | 95,581,650 | 55.6% |
| | Very low (already known or likely grassland/savanna) | 2,398,774 | 1.4% |
| ↓ Low | Lowest (not identified as historic grassland/savanna) | 52,005,114 | 30.3% |
| | *Area not evaluated for this indicator* | *18,545* | *<0.1%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Greenways & trails

This cultural resource indicator measures both the natural condition and connected length of greenways and trails to characterize the quality of the recreational experience. Natural condition is based on the amount of impervious surface surrounding the path. Connected length captures how far a person can go without leaving a dedicated path, based on common distances for walking, running, and biking. This indicator originates from OpenStreetMap data and the National Land Cover Database.




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- Mostly natural and connected for ≥40 km
- Mostly natural and connected for 5 to <40 km or partly natural and connected for ≥40 km
- Mostly natural and connected for 1.9 to <5 km, partly natural and connected for 5 to <40 km, or developed and connected for ≥40 km
- Mostly natural and connected for <1.9 km, partly natural and connected for 1.9 to <5 km, or developed and connected for 5 to <40 km
- Partly natural and connected for <1.9 km or developed and connected for 1.9 to <5 km
- Developed and connected for <1.9 km
- Sidewalk
- Not identified as a trail, sidewalk, or other path

*Table 9: Indicator values for greenways & trails within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

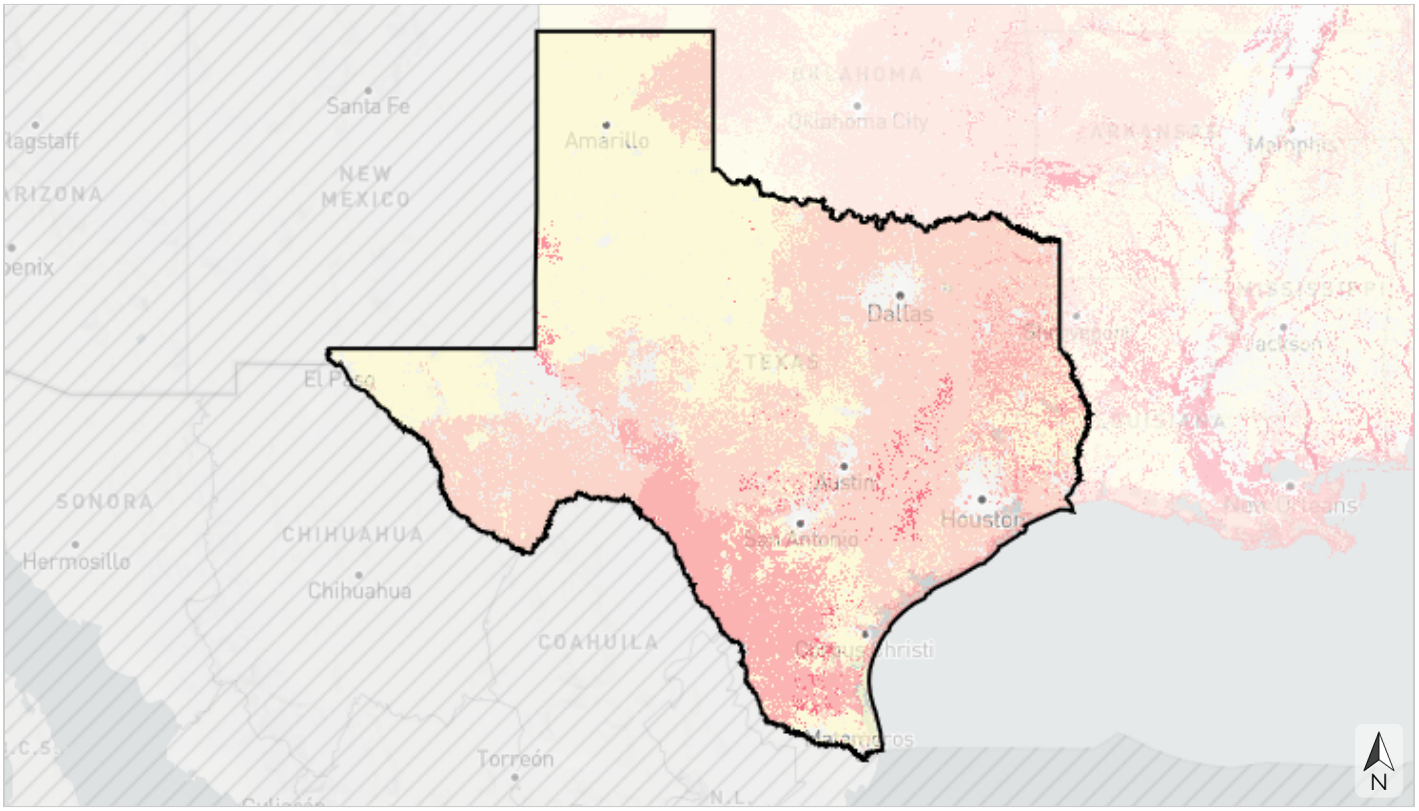
| | Indicator Values | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Mostly natural and connected for ≥40 km | 10,725 | <0.1% | |
| | Mostly natural and connected for 5 to <40 km or partly natural and connected for ≥40 km | 20,640 | <0.1% | |
| | Mostly natural and connected for 1.9 to <5 km, partly natural and connected for 5 to <40 km, or developed and connected for ≥40 km | 41,278 | <0.1% | |
| | Mostly natural and connected for <1.9 km, partly natural and connected for 1.9 to <5 km, or developed and connected for 5 to <40 km | 31,134 | <0.1% | ↑ In good condition |
| | Partly natural and connected for <1.9 km or developed and connected for 1.9 to <5 km | 18,180 | <0.1% | ↓ Not in good condition |
| | Developed and connected for <1.9 km | 28,223 | <0.1% | |
| | Sidewalk | 65,229 | <0.1% | |
| ↓ Low | Not identified as a trail, sidewalk, or other path | 171,630,447 | 99.8% | |
| | *Area not evaluated for this indicator* | *51,569* | *<0.1%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Imperiled amphibians & reptiles

This indicator identifies potential habitat to support amphibian and reptile Regional Species of Greatest Conservation Need (RSGCN). Using a rarity-weighted richness approach, it prioritizes places important for species that are restricted to smaller ranges, such as narrow endemics, which are therefore more vulnerable to habitat loss or change. RSGCN are regional priority species derived from the list of SGCN identified in Southeast State Wildlife Action Plans as most in need of conservation action. RSGCN were chosen based on consistent criteria, such as level of conservation concern, regional stewardship responsibility, and ecological significance. This indicator uses 190 species and subspecies models from the U.S. Geological Survey Gap Analysis Project that are based on known ranges, habitat relationships, and LANDFIRE landcover.




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**Importance for range-restricted species**

- Highest importance (top 5%)
- Very high importance (upper 5-10%)
- High importance (upper 10-20%)
- Medium importance (above average)
- Low importance (below average)
- Developed landcover or no potential habitat

*Table 10: Indicator values for imperiled amphibians & reptiles within Texas. A good condition threshold is not yet defined for this indicator.*

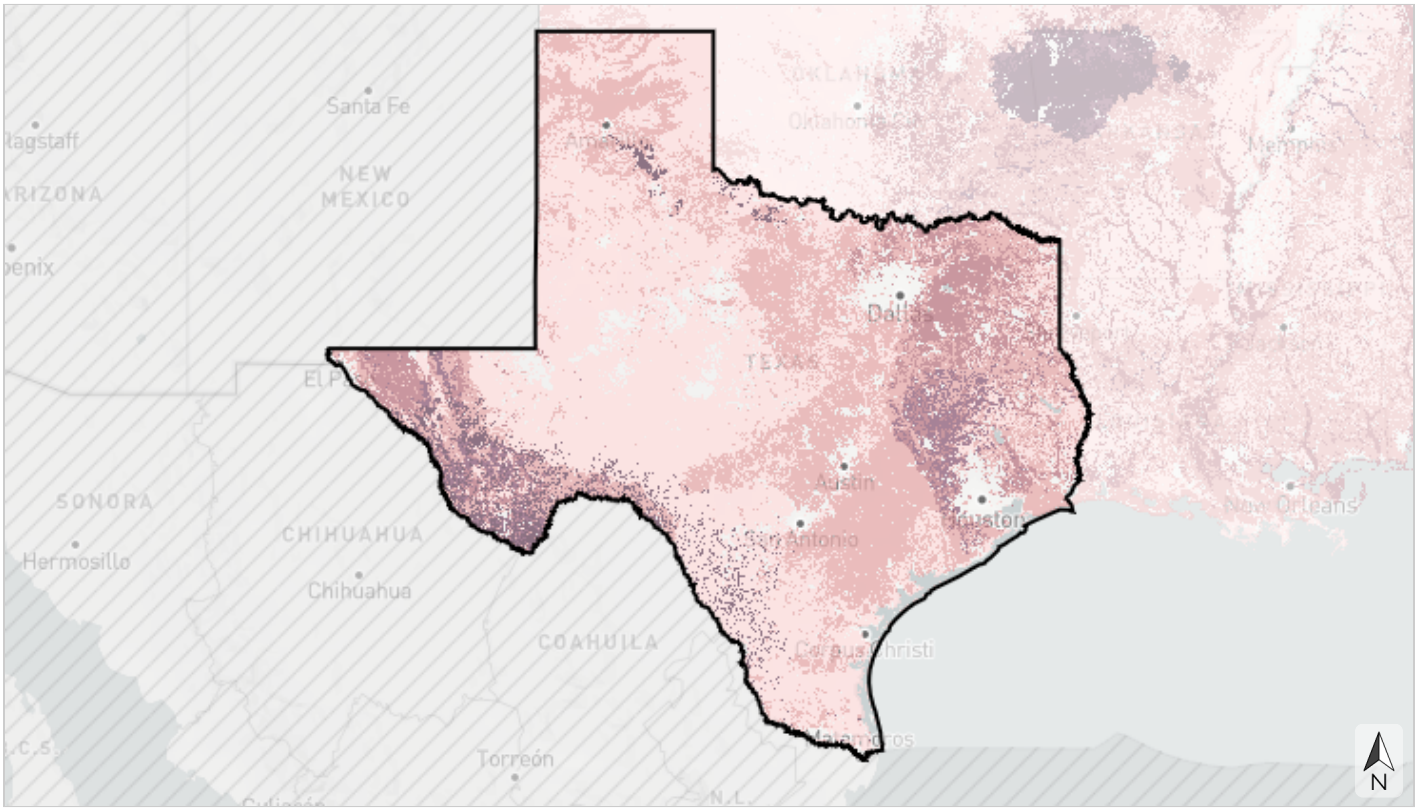
| | Indicator Values: Importance for range-restricted species | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Highest importance (top 5%) | 3,400,204 | 2.0% |
| | Very high importance (upper 5-10%) | 1,374,221 | 0.8% |
| | High importance (upper 10-20%) | 19,952,920 | 11.6% |
| | Medium importance (above average) | 61,661,821 | 35.9% |
| | Low importance (below average) | 63,599,104 | 37.0% |
| ↓ Low | Developed landcover or no potential habitat | 20,287,287 | 11.8% |
| | *Area not evaluated for this indicator* | *1,621,869* | *0.9%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Imperiled mammals

This indicator identifies potential habitat to support mammal Regional Species of Greatest Conservation Need (RSGCN). Using a rarity-weighted richness approach, it prioritizes places important for species that are restricted to smaller ranges, such as narrow endemics, which are therefore more vulnerable to habitat loss or change. RSGCN are regional priority species derived from the list of SGCN identified in Southeast State Wildlife Action Plans as most in need of conservation action. RSGCN were chosen based on consistent criteria, such as level of conservation concern, regional stewardship responsibility, and ecological significance. This indicator uses 58 species models from the U.S. Geological Survey Gap Analysis Project that are based on known ranges, habitat relationships, and LANDFIRE landcover.




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**Importance for range-restricted species**

- Highest importance (top 5%)
- Very high importance (upper 5-10%)
- High importance (upper 10-20%)
- Medium importance (above average)
- Low importance (below average)
- Developed landcover or no potential habitat

*Table 11: Indicator values for imperiled mammals within Texas. A good condition threshold is not yet defined for this indicator.*

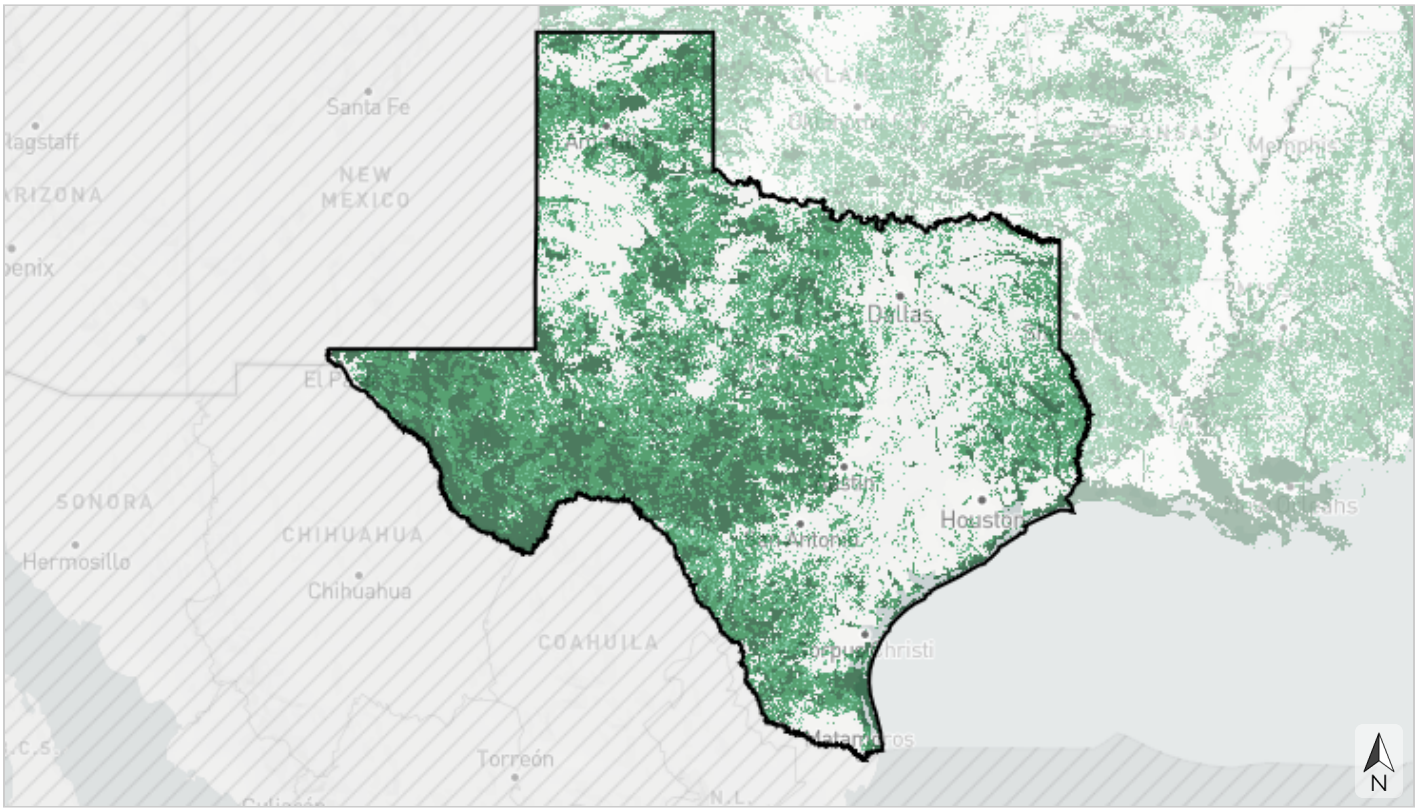
| | Indicator Values: Importance for range-restricted species | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Highest importance (top 5%) | 6,027,173 | 3.5% |
| | Very high importance (upper 5-10%) | 5,634,908 | 3.3% |
| | High importance (upper 10-20%) | 12,647,509 | 7.4% |
| | Medium importance (above average) | 50,569,386 | 29.4% |
| | Low importance (below average) | 75,450,446 | 43.9% |
| ↓ Low | Developed landcover or no potential habitat | 19,946,135 | 11.6% |
| | *Area not evaluated for this indicator* | *1,621,869* | *0.9%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Intact habitat cores

This indicator represents the size of large, unfragmented patches of natural habitat. It identifies minimally disturbed natural areas at least 100 acres in size and greater than 200 meters wide. Large areas of intact natural habitat are important for many wildlife species, including reptiles and amphibians, birds, and large mammals. This indicator originates from Esri's green infrastructure data.




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- Large core (>10,000 acres)
- Medium core (>1,000-10,000 acres)
- Small core (>100-1,000 acres)
- Not a core

*Table 12: Indicator values for intact habitat cores within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

| | Indicator Values | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Large core (>10,000 acres) | 31,116,399 | 18.1% | |
| | Medium core (>1,000-10,000 acres) | 44,602,239 | 25.9% | |
| | Small core (>100-1,000 acres) | 20,494,756 | 11.9% | ↑ In good condition |
| ↓ Low | Not a core | 75,645,108 | 44.0% | ↓ Not in good condition |
| | *Area not evaluated for this indicator* | *38,923* | *<0.1%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Landscape condition

This indicator represents natural areas with limited human alteration while also considering the naturalness of the surrounding landscape. Examples of human alteration include urban development and intense agricultural use. The degree of naturalness across the landscape is a key ecological condition for sustaining species and ecosystem services that are sensitive to habitat fragmentation at multiple scales. This indicator combines data from multiple sources, including the National Land Cover Database; various data on grasslands, mines, quarries, and reservoirs; ideas from the Florida Critical Lands and Waters Identification Project's approach for evaluating land use intensity; and more.




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- Very natural landscape
- Natural landscape
- Mostly natural landscape
- Partly natural landscape
- Altered landscape
- Heavily altered landscape

*Table 13: Indicator values for landscape condition within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

| | Indicator Values | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Very natural landscape | 41,858,192 | 24.4% | |
| | Natural landscape | 42,783,216 | 24.9% | |
| | Mostly natural landscape | 34,126,949 | 19.9% | ↑ In good condition |
| | Partly natural landscape | 44,679,226 | 26.0% | ↓ Not in good condition |
| | Altered landscape | 5,170,998 | 3.0% | |
| ↓ Low | Heavily altered landscape | 1,657,867 | 1.0% | |
| | *Area not evaluated for this indicator* | *1,620,978* | *0.9%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Playas

This indicator represents the condition and location of playas, which are round, shallow depressions found primarily in the western Great Plains that serve as temporary wetlands by collecting water from rainfall and runoff. It defines a healthy playa as one that is not farmed, hydrologically modified, within a wind farm, or impacted by sediment accumulation due to agriculture. It also considers the increased benefits to wildlife provided by clusters of nearby playas, compared to more sparsely distributed playas. Playas play a critical role in recharging the Ogallala aquifer and provide habitat and food for birds and other animals. This indicator originates from the Playa Lakes Joint Venture probable playas dataset.




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- Healthy playa and part of a larger cluster
- Healthy playa
- Other playa
- Not identified as a playa

*Table 14: Indicator values for playas within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

| | Indicator Values | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Healthy playa and part of a larger cluster | 139,261 | <0.1% | |
| | Healthy playa | 4,004 | <0.1% | ↑ In good condition |
| | Other playa | 249,050 | 0.1% | ↓ Not in good condition |
| ↓ Low | Not identified as a playa | 47,766,454 | 27.8% | |
| | *Area not evaluated for this indicator* | *123,738,656* | *72.0%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Potential access to parks

This cultural resource indicator prioritizes places to create new parks that would fill gaps in access to open space within socially vulnerable communities in urban areas. It identifies areas where residents currently lack access to parks within a 10-minute walk (accounting for walkable road networks and access barriers like highways and fences), then prioritizes based on park need using demographic and environmental metrics. Parks help improve public health, foster a conservation ethic by providing opportunities for people to connect with nature, and support critical ecosystem services. This indicator originates from the Trust for Public Land's ParkServe park priority areas and the Center for Disease Control's Social Vulnerability Index.




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**Priority for a new park to serve people who lack nearby park access**

- Very high priority
- High priority
- Moderate priority
- Not identified as a priority (within urban areas)

*Table 15: Indicator values for potential access to parks within Texas. A good condition threshold is not yet defined for this indicator.*

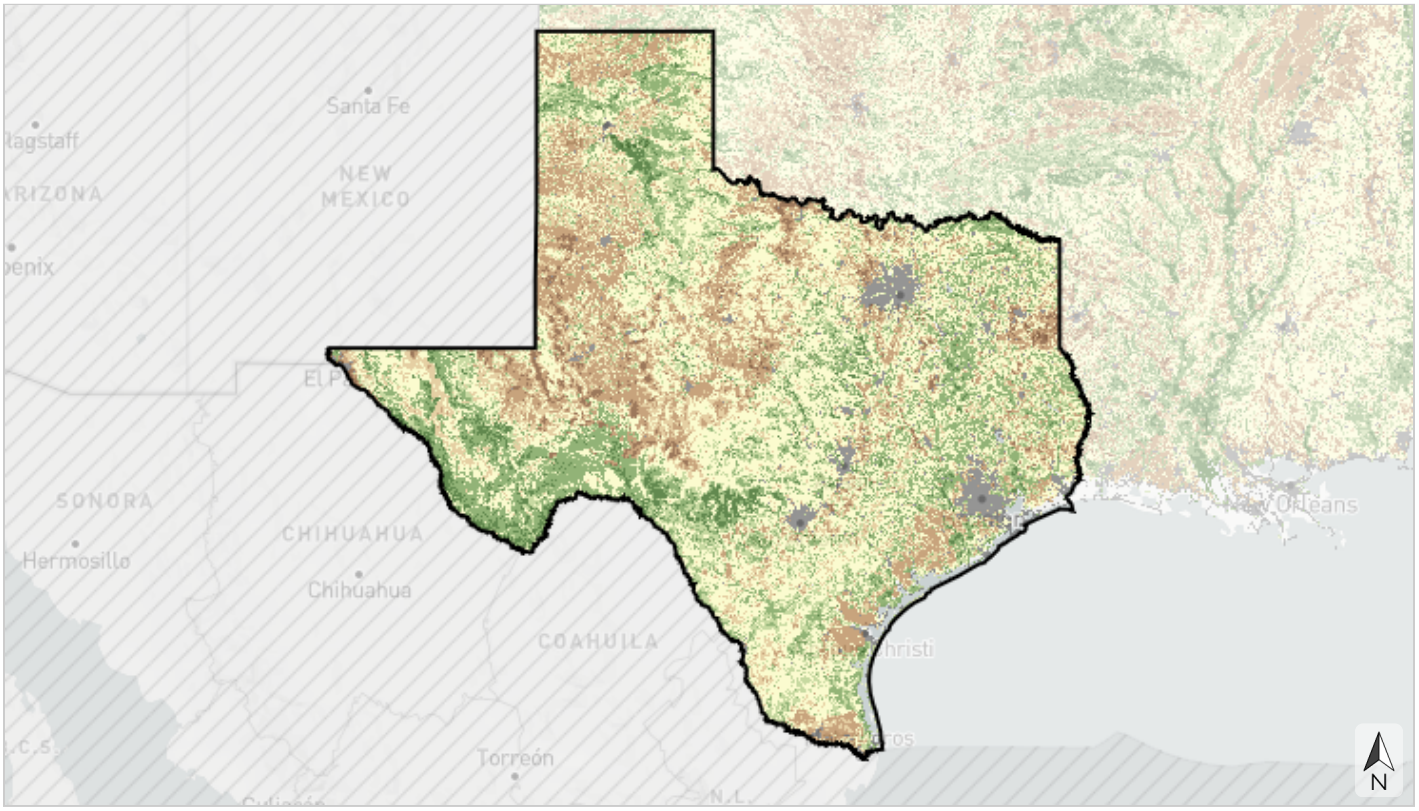
| | Indicator Values: Priority for a new park to serve people who lack nearby park access | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Very high priority | 281,370 | 0.2% |
| | High priority | 414,646 | 0.2% |
| | Moderate priority | 804,898 | 0.5% |
| ↓ Low | Not identified as a priority (within urban areas) | 168,422,819 | 98.0% |
| | *Area not evaluated for this indicator* | *1,973,692* | *1.1%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Resilient terrestrial sites

This indicator depicts an area's capacity to maintain species diversity and ecosystem function in the face of climate change. It measures two factors that influence resilience. The first, landscape diversity, reflects the number of microhabitats and climatic gradients created by topography, elevation, and hydrology. The second, local connectedness, reflects the degree of habitat fragmentation and strength of barriers to species movement. Highly resilient sites contain many different habitat niches that support biodiversity, and allow species to move freely through the landscape to find suitable microclimates as the climate changes. This indicator originates from The Nature Conservancy's Resilient Land data.




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- Most resilient
- More resilient
- Slightly more resilient
- Average/median resilience
- Slightly less resilient
- Less resilient
- Least resilient
- Developed

*Table 16: Indicator values for resilient terrestrial sites within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Most resilient | 5,952,031 | 3.5% |
| | More resilient | 26,536,347 | 15.4% |
| | Slightly more resilient | 25,359,363 | 14.8% |
| | Average/median resilience | 53,370,991 | 31.0% |
| | Slightly less resilient | 24,270,669 | 14.1% |
| | Less resilient | 21,631,302 | 12.6% |
| | Least resilient | 3,301,617 | 1.9% |
| ↓ Low | Developed | 5,488,958 | 3.2% |
| | *Area not evaluated for this indicator* | *5,986,147* | *3.5%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# River cane restoration

This indicator represents priority places for river cane restoration and co-stewardship near the lands of federally recognized Tribes. River cane is a bamboo species native to the Southeast. Historically, it was abundant and widespread, forming dense patches called canebrakes that could stretch for miles within the floodplain. Today, it has been reduced to less than 2% of its former extent. River cane not only provides significant habitat value and ecosystem services, but many Southeastern Native American Tribes consider it a cultural keystone species important to the continuity of traditional lifeways. This indicator combines data from multiple sources, including Natural Resources Conservation Service soils data, the Environmental Protection Agency's estimated floodplain, lands of federally recognized Tribes lands from the U.S. Census Bureau, and more.




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- Potential river cane restoration/management area on Tribal land
- Potential river cane restoration/management area on protected land near Tribal lands
- Potential river cane restoration/management area near Tribal lands
- Potential river cane restoration/management area on protected land
- Potential river cane restoration/management area
- Not identified as a rivercane restoration/management area

*Table 17: Indicator values for river cane restoration within Texas. A good condition threshold is not yet defined for this indicator.*

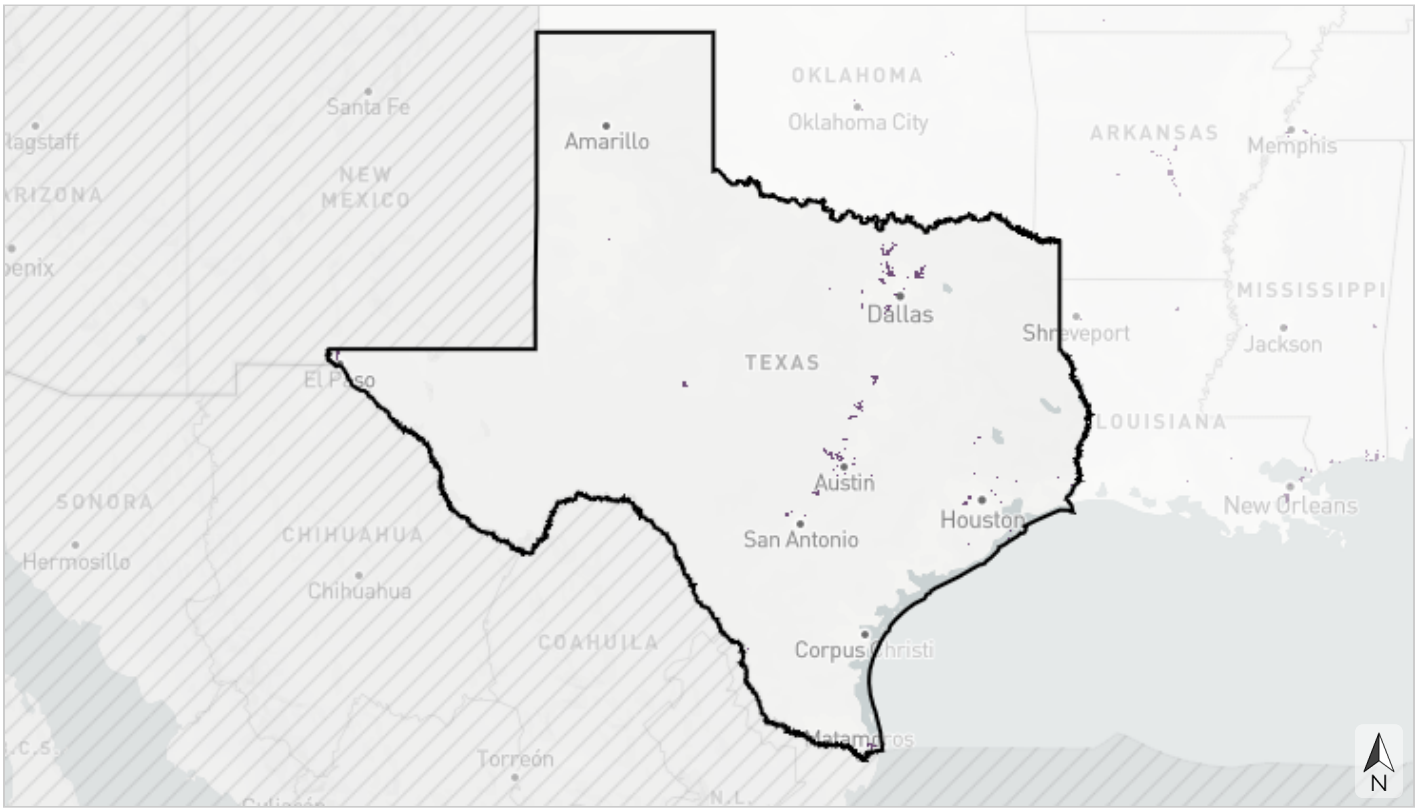
| | Indicator Values | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Potential river cane restoration/management area on Tribal land | 2,597 | <0.1% |
| | Potential river cane restoration/management area on protected land near Tribal lands | 327,696 | 0.2% |
| | Potential river cane restoration/management area near Tribal lands | 5,191,105 | 3.0% |
| | Potential river cane restoration/management area on protected land | 876,986 | 0.5% |
| | Potential river cane restoration/management area | 15,043,433 | 8.8% |
| ↓ Low | Not identified as a rivercane restoration/management area | 147,959,270 | 86.1% |
| | *Area not evaluated for this indicator* | *2,496,339* | *1.5%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# Urban park size

This cultural resource indicator measures the size of parks larger than 5 acres in the urban environment. Protected natural areas in urban environments provide urban residents a nearby place to connect with nature, and offer refugia for some species. This indicator complements the equitable access to potential parks indicator by capturing the value of existing parks. It originates from the Protected Areas Database of the United States, U.S. Census Bureau urban areas, and the National Land Cover Database.




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- 75+ acre urban park
- 50 to <75 acre urban park
- 30 to <50 acre urban park
- 10 to <30 acre urban park
- 5 to <10 acre urban park
- <5 acre urban park
- Not identified as an urban park

*Table 18: Indicator values for urban park size within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | 75+ acre urban park | 611,553 | 0.4% |
| | 50 to <75 acre urban park | 19,438 | <0.1% |
| | 30 to <50 acre urban park | 24,217 | <0.1% |
| | 10 to <30 acre urban park | 37,826 | <0.1% |
| | 5 to <10 acre urban park | 16,537 | <0.1% |
| | <5 acre urban park | 16,614 | <0.1% |
| ↓ Low | Not identified as an urban park | 170,746,781 | 99.3% |
| | *Area not evaluated for this indicator* | *424,459* | *0.2%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# West Coastal Plain & Ouachitas forested wetland birds

This indicator is an index of habitat suitability for five forested wetland bird species (Acadian flycatcher, Kentucky warbler, yellow-throated warbler, prothonotary warbler, red-shouldered hawk) within bottomland hardwood forests and riparian areas in the West Gulf Coastal Plain/Ouachitas (WGCPO) Bird Conservation Region. It uses metrics like patch size, dispersal distance, and distance to water to assess the potential for habitat to support sustainable populations of these birds. This indicator originates from the Lower Mississippi Valley Joint Venture's forested wetland decision support model for the WGCPO region.




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**Habitat suitability for forested wetland bird umbrella species**

- High habitat suitability (score >80)
- Medium-high habitat suitability (score >60-80)
- Medium habitat suitability (score >40-60)
- Medium-low habitat suitability (score >20-40)
- Low habitat suitability (score >0-20)
- Not suitable (score = 0)

*Table 19: Indicator values for West Coastal Plain & Ouachitas forested wetland birds within Texas. A good condition threshold is not yet defined for this indicator.*

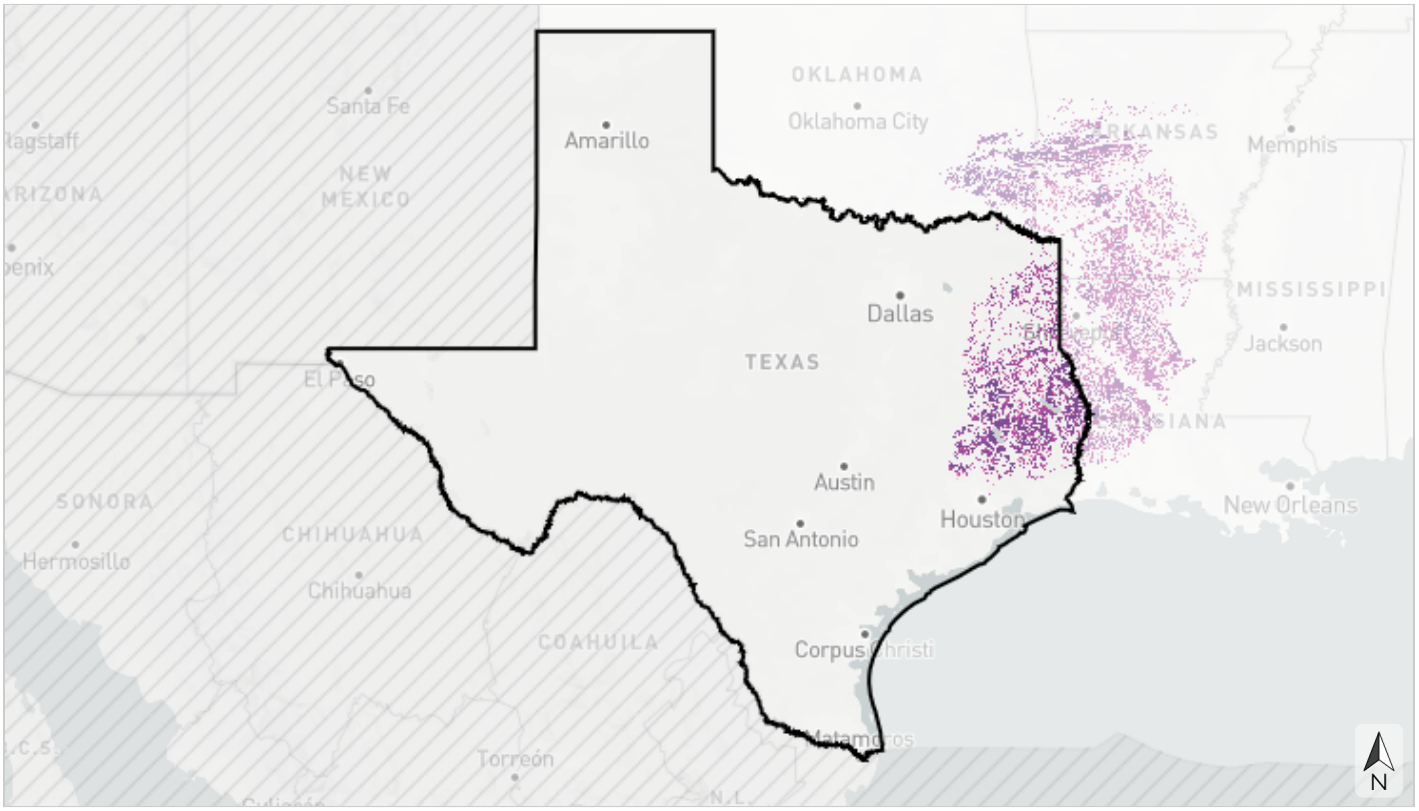
| | Indicator Values: Habitat suitability for forested wetland bird umbrella species | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | High habitat suitability (score >80) | 472,163 | 0.3% |
| | Medium-high habitat suitability (score >60-80) | 318,899 | 0.2% |
| | Medium habitat suitability (score >40-60) | 361,743 | 0.2% |
| | Medium-low habitat suitability (score >20-40) | 566,631 | 0.3% |
| | Low habitat suitability (score >0-20) | 553,694 | 0.3% |
| ↓ Low | Not suitable (score = 0) | 13,606,700 | 7.9% |
| | *Area not evaluated for this indicator* | *156,017,595* | *90.8%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# West Coastal Plain & Ouachitas open pine birds

This indicator identifies areas with pine trees that, if managed for open condition, could support a population of three umbrella bird species (brown-headed nuthatch, Bachman's sparrow, red-cockaded woodpecker). It evaluates potential habitat in the West Gulf Coastal Plain/Ouachitas (WGCPO) Bird Conservation Region based on each species' habitat needs and population dynamics, prioritizing opportunities to restore and manage habitat to benefit open pine birds. Final scores reflect both the selectiveness of the species and whether an area meets the habitat requirements through one large patch, or clusters of smaller patches in sufficiently close proximity for breeding pairs to disperse. This indicator updates the Lower Mississippi Valley Joint Venture's open pine decision support model for the WGCPO region.




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**Ability of pine patch to support a population of umbrella bird species if managed in open condition**

- Large enough to support a population of all 3 species
- Large enough to support a population of 2 species
- Large enough to a population of 1 species
- Part of a cluster of nearby patches able to support a population of all 3 species
- Part of a cluster of nearby patches able to support a population of 2 species
- Part of a cluster of nearby patches able to support a population of 1 species
- Pine patch too small and isolated to support a population of any species or not an upland pine patch

*Table 20: Indicator values for West Coastal Plain & Ouachitas open pine birds within Texas. A good condition threshold is not yet defined for this indicator.*

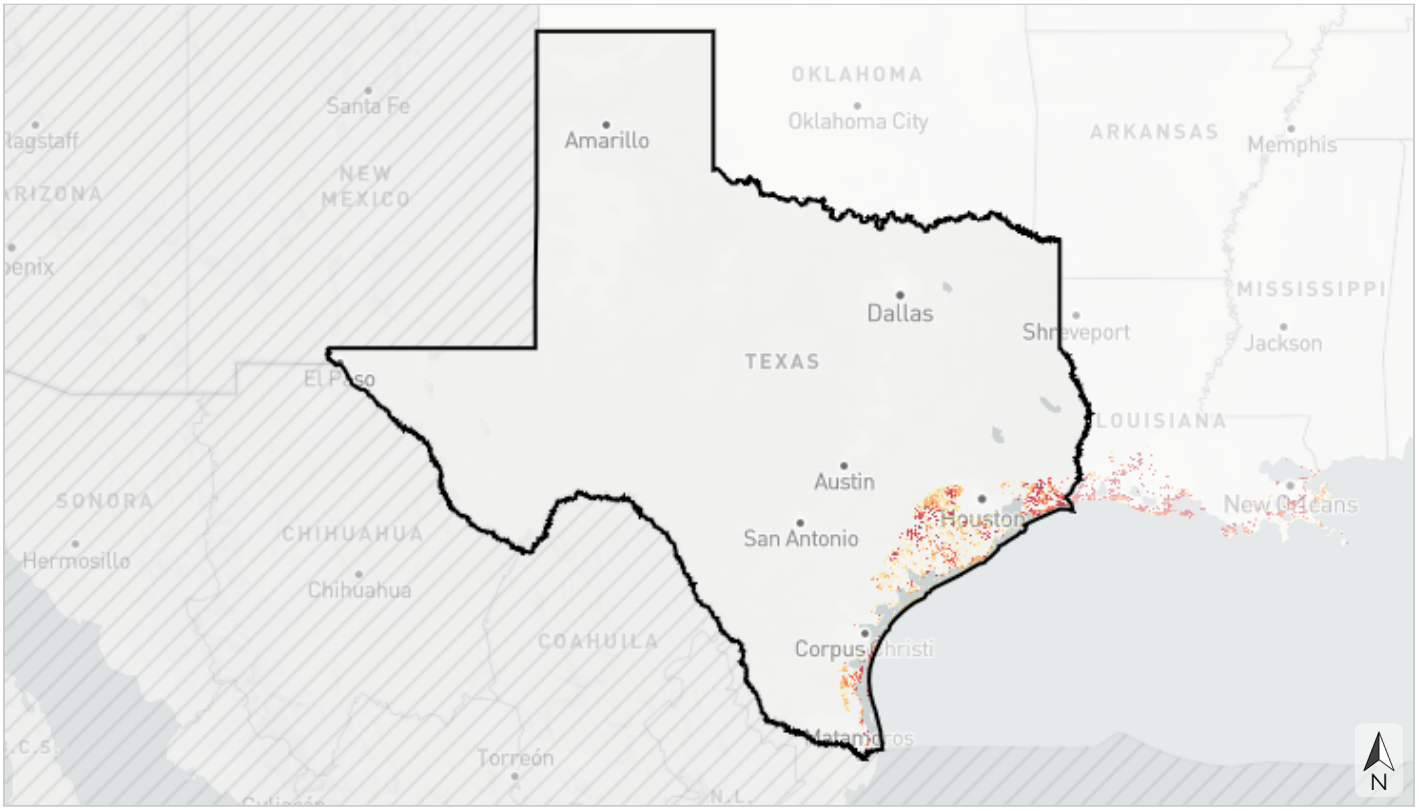
| | Indicator Values: Ability of pine patch to support a population of umbrella bird species if managed in open condition | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Large enough to support a population of all 3 species | 1,589,538 | 0.9% |
| | Large enough to support a population of 2 species | 2,163,307 | 1.3% |
| | Large enough to a population of 1 species | 559,981 | 0.3% |
| | Part of a cluster of nearby patches able to support a population of all 3 species | 415,274 | 0.2% |
| | Part of a cluster of nearby patches able to support a population of 2 species | 1,057,850 | 0.6% |
| | Part of a cluster of nearby patches able to support a population of 1 species | 5,709 | <0.1% |
| ↓ Low | Pine patch too small and isolated to support a population of any species or not an upland pine patch | 10,087,885 | 5.9% |
| | *Area not evaluated for this indicator* | *156,017,881* | *90.8%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Terrestrial

# West Gulf Coast mottled duck nesting

This indicator depicts marshes and grasslands along the coast of Louisiana and Texas that are important for mottled duck nesting, based on key biological parameters such as patch size, land cover type, and distance to brood rearing habitat. As a non-migratory bird endemic to the Gulf coast, mottled ducks serve as good indicators of coastal marsh health and function. Urban growth, agricultural development, and hydrologic changes due to human alteration and climate change have caused significant mottled duck habitat loss and population declines. This indicator originates from a mottled duck decision support tool developed by the Gulf Coast Joint Venture.




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**Percentile of suitable mottled duck nesting habitat**

- 90th-100th percentile
- 80th-90th percentile
- 70th-80th percentile
- 60th-70th percentile
- 50th-60th percentile
- 40th-50th percentile
- 30th-40th percentile
- 20th-30th percentile
- 10th-20th percentile
- 0-10th percentile
- Not identified as suitable (within TX and LA)

*Table 21: Indicator values for West Gulf Coast mottled duck nesting within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values: Percentile of suitable mottled duck nesting habitat | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | 90th-100th percentile | 150,896 | <0.1% |
| | 80th-90th percentile | 230,369 | 0.1% |
| | 70th-80th percentile | 265,008 | 0.2% |
| | 60th-70th percentile | 281,124 | 0.2% |
| | 50th-60th percentile | 307,163 | 0.2% |
| | 40th-50th percentile | 374,127 | 0.2% |
| | 30th-40th percentile | 398,233 | 0.2% |
| | 20th-30th percentile | 405,638 | 0.2% |
| | 10th-20th percentile | 405,367 | 0.2% |
| | 0-10th percentile | 359,904 | 0.2% |
| ↓ Low | Not identified as suitable (within TX and LA) | 7,717,798 | 4.5% |
| | *Area not evaluated for this indicator* | *161,001,797* | *93.7%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

# Floodplain inundation

This indicator uses the frequency of inundation to measure the hydrologic function of the floodplain. In a natural floodplain, water can move freely from mainstem rivers and lakes onto different parts of the floodplain. This cycle of frequent, but not persistent, flooding supports many aquatic species and habitats, like fish, waterfowl and swamp forests. It also provides essential ecosystem services such as nutrient retention and downstream flood control. When parts of the floodplain stay constantly wet or dry, this may signal negative impacts from structures like dikes, dams, and levees, which disconnect waterbodies from their adjacent floodplains. This indicator is based on Sentinel-2 satellite imagery and methods developed by Yvonne Allen of the U.S. Fish and Wildlife Service.




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**Frequency of inundation within the floodplain**

- Frequent inundation (flooded in 21-90% of days with available data)
- Regular inundation (flooded in 6-20% of days with available data)
- Occasional inundation (flooded in 2-5% of days with available data)
- Persistent inundation (flooded in 91-100% of days with available data)
- No detected inundation (flooded in 0% of days with available data)
- Not identified as a floodplain

*Table 22: Indicator values for floodplain inundation within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

| | Indicator Values: Frequency of inundation within the floodplain | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Frequent inundation (flooded in 21-90% of days with available data) | 1,116,295 | 0.6% | |
| | Regular inundation (flooded in 6-20% of days with available data) | 1,090,673 | 0.6% | ↑ In good condition |
| | Occasional inundation (flooded in 2-5% of days with available data) | 1,241,564 | 0.7% | ↓ Not in good condition |
| | Persistent inundation (flooded in 91-100% of days with available data) | 2,526,739 | 1.5% | |
| | No detected inundation (flooded in 0% of days with available data) | 17,977,118 | 10.5% | |
| ↓ Low | Not identified as a floodplain | 147,081,793 | 85.6% | |
| | *Area not evaluated for this indicator* | *863,243* | *0.5%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

# Imperiled aquatic species

This indicator measures the number of aquatic animal Regional Species of Greatest Conservation Need (RSGCN) observed within each 12-digit HUC subwatershed, including fish, mussels, snails, crayfish, and amphibians. It captures patterns of rare and endemic aquatic species diversity. RSGCN are regional priority species derived from the list of SGCN identified in Southeast State Wildlife Action Plans as most in need of need of conservation action. RSGCN were chosen based on consistent criteria, such as level of conservation concern, regional stewardship responsibility, and ecological significance. This indicator originates from state Natural Heritage Program data collected by the Southeast Aquatic Resources Partnership and applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).




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**Number of aquatic animal Regional Species of Greatest Conservation Need (RSGCN) observed**

- 8+ species
- 7 species
- 6 species
- 5 species
- 4 species
- 3 species
- 2 species
- 1 species
- 0 species
- Not identified as a floodplain

*Table 23: Indicator values for imperiled aquatic species within Texas. A good condition threshold is not yet defined for this indicator.*

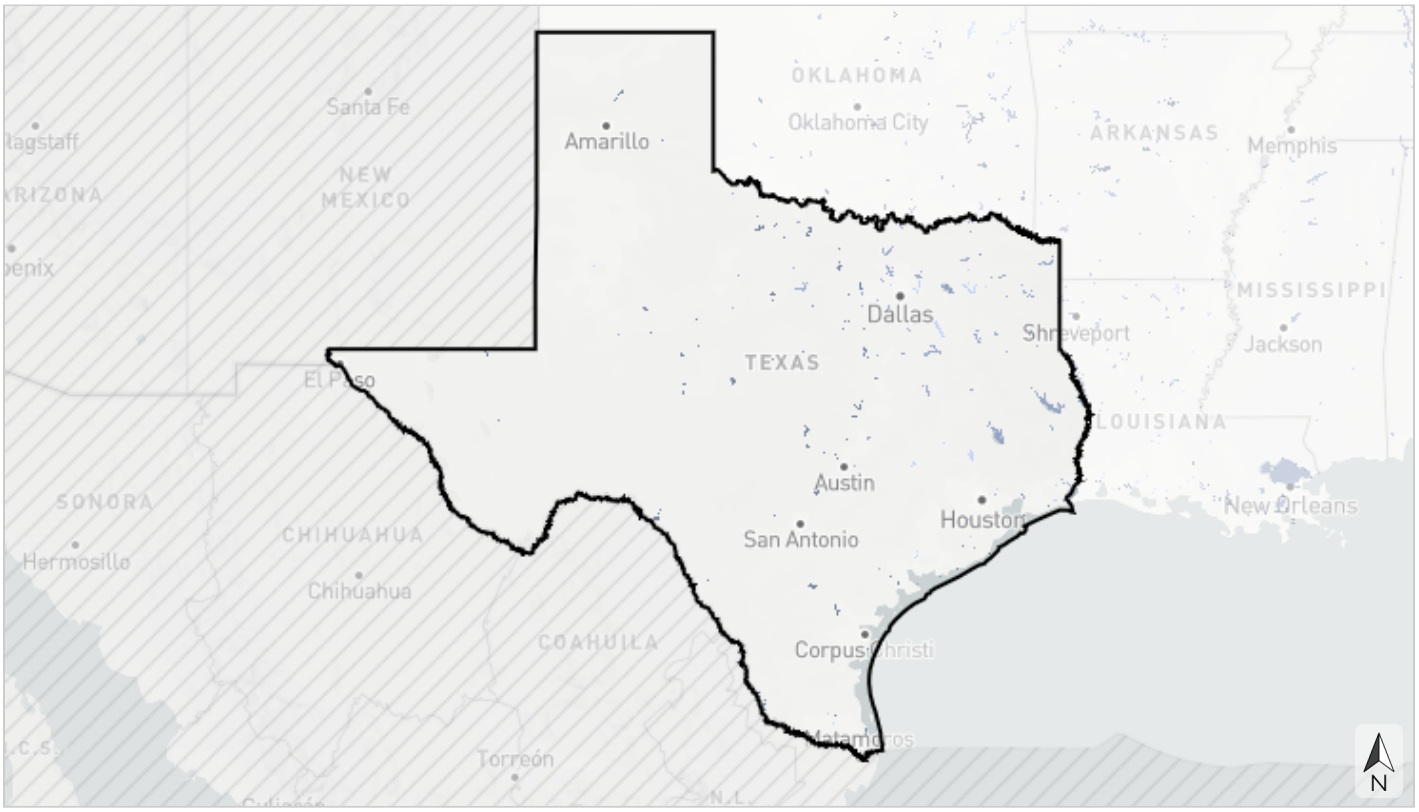
| | Indicator Values: Number of aquatic animal Regional Species of Greatest Conservation Need (RSGCN) observed | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | 8+ species | 221,955 | 0.1% |
| | 7 species | 133,143 | <0.1% |
| | 6 species | 205,787 | 0.1% |
| | 5 species | 430,452 | 0.3% |
| | 4 species | 468,540 | 0.3% |
| | 3 species | 1,228,067 | 0.7% |
| | 2 species | 1,644,315 | 1.0% |
| | 1 species | 2,471,257 | 1.4% |
| | 0 species | 15,538,142 | 9.0% |
| ↓ Low | Not identified as a floodplain | 147,065,874 | 85.6% |
| | *Area not evaluated for this indicator* | *2,489,893* | *1.4%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

# Lakes & reservoirs

This indicator assesses the condition of lakes and reservoirs based on the amount of natural landcover within the upstream watershed. Higher scores go to lakes and reservoirs with less surrounding urban and agricultural development (including intensive forest management for timber production), which negatively impact drinking water quality and quantity and fish habitat. This indicator uses waterbody and watershed data from the LAGOS-LOCUS dataset, as well as landcover from the National Land Cover Database and LANDFIRE.




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- Lake or reservoir with low disturbance (0-25%) in upstream watershed
- Lake or reservoir with medium disturbance (>25-60%) in upstream watershed
- Lake or reservoir with high disturbance (>60%) in upstream watershed
- Not identified as a lake or reservoir

*Table 24: Indicator values for lakes & reservoirs within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

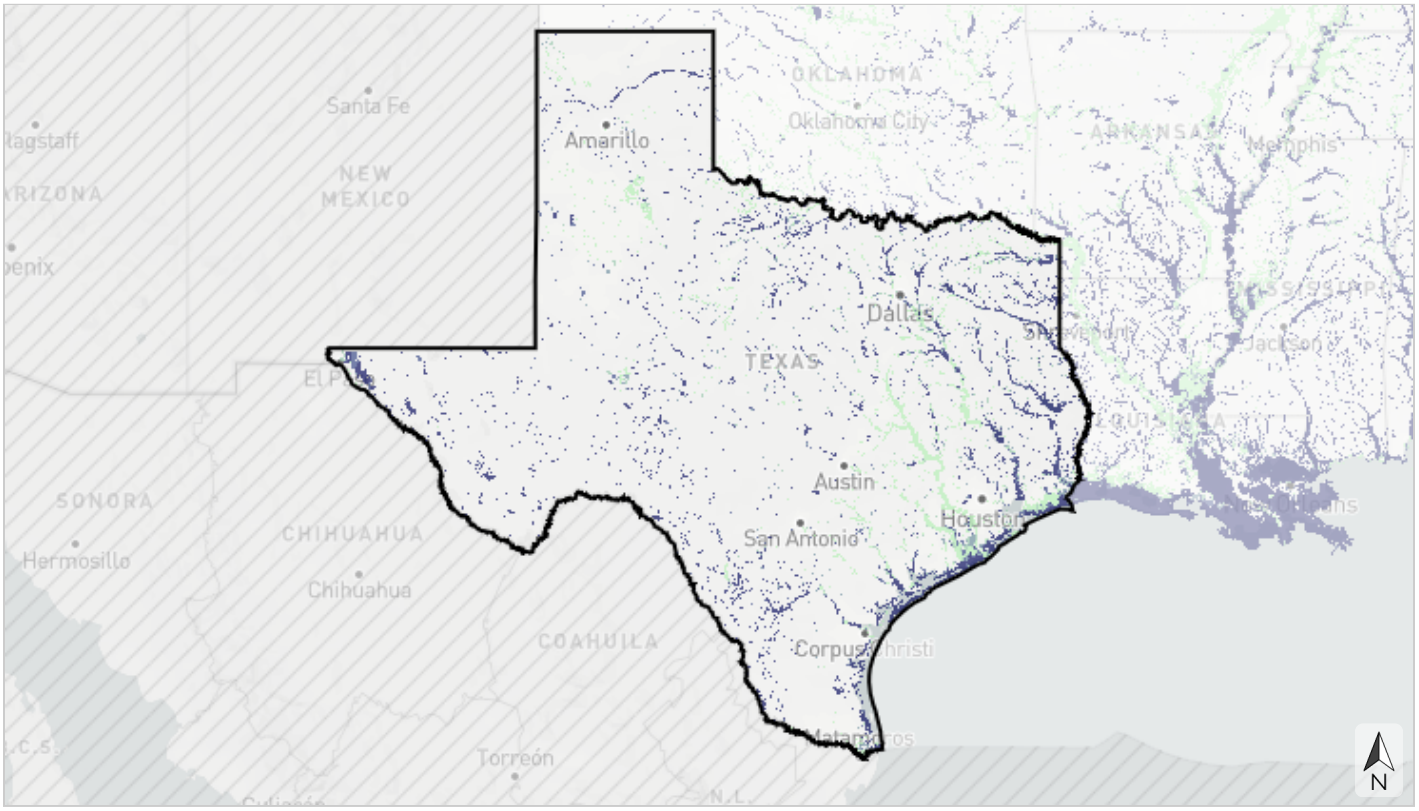
| | Indicator Values | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Lake or reservoir with low disturbance (0-25%) in upstream watershed | 632,299 | 0.4% | ↑ In good condition |
| | Lake or reservoir with medium disturbance (>25-60%) in upstream watershed | 876,191 | 0.5% | ↓ Not in good condition |
| | Lake or reservoir with high disturbance (>60%) in upstream watershed | 344,411 | 0.2% | |
| ↓ Low | Not identified as a lake or reservoir | 170,005,601 | 98.9% | |
| | *Area not evaluated for this indicator* | *38,923* | *<0.1%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

# Natural landcover in floodplains

This indicator measures the amount of natural landcover in the estimated floodplain of rivers and streams within each catchment. It assesses the stream channel and its surrounding riparian buffer, measuring the percent of unaltered habitat like forests, wetlands, or open water (rather than agriculture or development). Intact vegetated buffers within the floodplain of rivers and streams provide aquatic habitat, improve water quality, reduce erosion and flooding, recharge groundwater, and more. This indicator originates from the National Land Cover Database and applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).




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**Percent natural landcover within the estimated floodplain, by catchment**

- >90% natural landcover
- >80-90% natural landcover
- >70-80% natural landcover
- >60-70% natural landcover
- ≤60% natural landcover
- Not identified as a floodplain

*Table 25: Indicator values for natural landcover in floodplains within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

| | Indicator Values: Percent natural landcover within the estimated floodplain, by catchment | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | >90% natural landcover | 11,519,379 | 6.7% | |
| | >80-90% natural landcover | 2,562,670 | 1.5% | ↑ In good condition |
| | >70-80% natural landcover | 1,833,007 | 1.1% | ↓ Not in good condition |
| | >60-70% natural landcover | 1,375,049 | 0.8% | |
| | ≤60% natural landcover | 5,051,553 | 2.9% | |
| ↓ Low | Not identified as a floodplain | 147,081,793 | 85.6% | |
| | *Area not evaluated for this indicator* | *2,473,974* | *1.4%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

# Network complexity

This indicator depicts the number of connected stream size classes in a river network between dams or waterfalls. River networks with a variety of connected stream classes help retain aquatic biodiversity in a changing climate by allowing species to access climate refugia and move between habitats. This indicator originates from the Southeast Aquatic Resources Partnership and applies to the Environmental Protection Agency's estimated floodplain, which spatially defines areas estimated to be inundated by a 100-year flood (also known as the 1% annual chance flood).




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**Number of connected stream size classes**

- 7 size classes
- 6 size classes
- 5 size classes
- 4 size classes
- 3 size classes
- 2 size classes
- 1 size class
- Not identified as a floodplain

*Table 26: Indicator values for network complexity within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

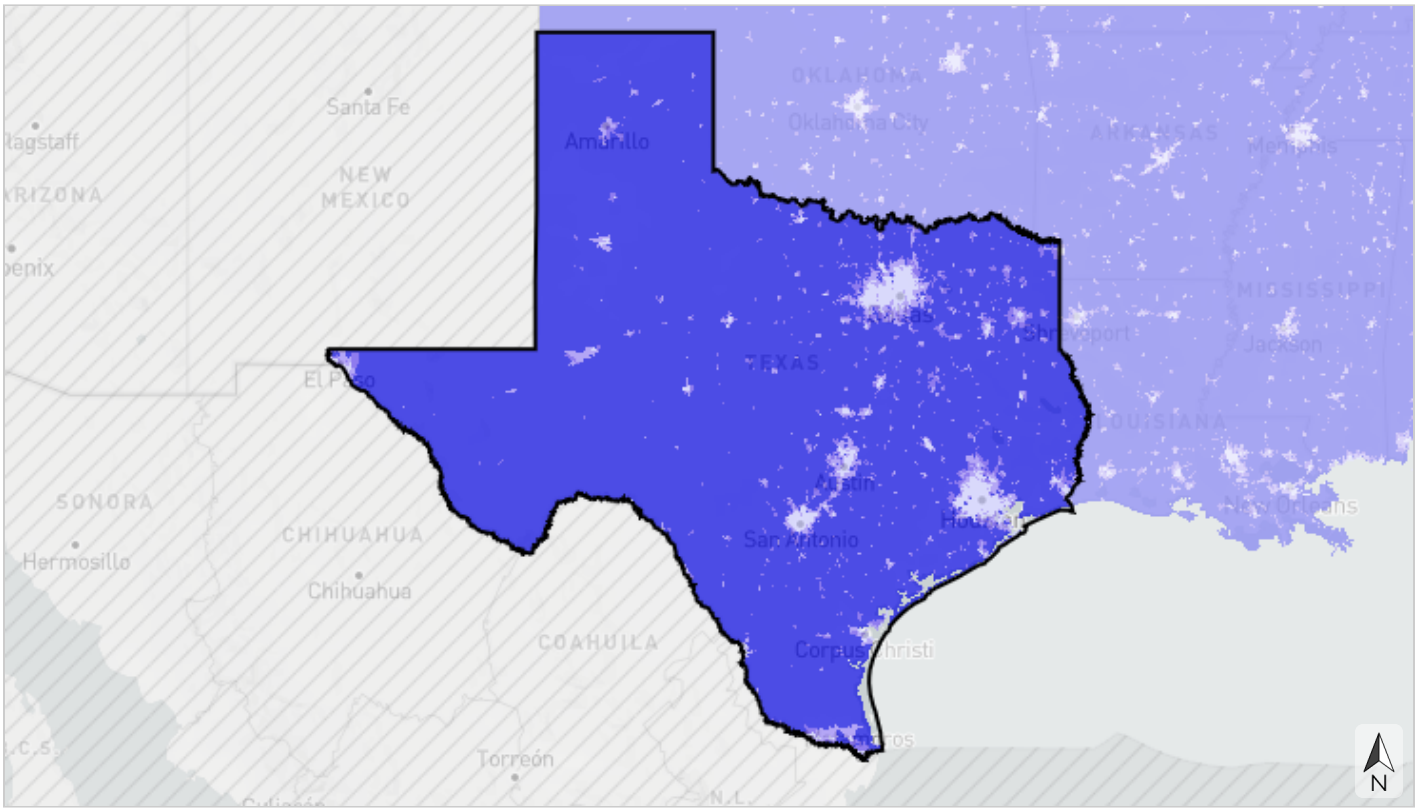
| | Indicator Values: Number of connected stream size classes | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | 7 size classes | 3,727,060 | 2.2% | |
| | 6 size classes | 3,591,382 | 2.1% | |
| | 5 size classes | 4,472,240 | 2.6% | |
| | 4 size classes | 2,657,194 | 1.5% | ↑ In good condition |
| | 3 size classes | 2,637,739 | 1.5% | ↓ Not in good condition |
| | 2 size classes | 2,839,417 | 1.7% | |
| | 1 size class | 2,058,003 | 1.2% | |
| ↓ Low | Not identified as a floodplain | 147,107,829 | 85.6% | |
| | *Area not evaluated for this indicator* | *2,806,560* | *1.6%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Freshwater

# Permeable surface

This indicator measures the average percent of non-impervious cover within each catchment. High levels of impervious surface degrade water quality and alter freshwater flow, impacting both aquatic species communities and ecosystem services for people, like the availability of clean drinking water. This indicator originates from the National Land Cover Database.




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**Percent of catchment permeable**

- >95% permeable (likely high water quality and supporting most sensitive aquatic species)
- >90-95% permeable (likely declining water quality and supporting most aquatic species)
- >70-90% permeable (likely degraded water quality and not supporting many aquatic species)
- ≤70% permeable (likely degraded instream flow, water quality, and aquatic species communities)

*Table 27: Indicator values for permeable surface within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

| | Indicator Values: Percent of catchment permeable | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | >95% permeable (likely high water quality and supporting most sensitive aquatic species) | 157,051,837 | 91.4% | ↑ In good condition |
| | >90-95% permeable (likely declining water quality and supporting most aquatic species) | 4,516,404 | 2.6% | ↓ Not in good condition |
| | >70-90% permeable (likely degraded water quality and not supporting many aquatic species) | 4,790,791 | 2.8% | |
| ↓ Low | ≤70% permeable (likely degraded instream flow, water quality, and aquatic species communities) | 3,045,088 | 1.8% | |
| | *Area not evaluated for this indicator* | *2,493,305* | *1.5%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Coastal shoreline condition

This indicator assesses shoreline condition based on the presence of hardened structures like jetties, groins, and riprap, as well as other human development. By restricting the natural movement of sediment, shoreline armoring increases erosion, prevents the inland migration of coastal ecosystems in response to sea level rise, and degrades habitat for birds, sea turtles, fish, plants, and other species both on and offshore. Natural shorelines in harder-to-develop coastal areas receive the highest shoreline condition scores, while hardened shorelines receive the lowest scores. This indicator originates from the National Oceanic and Atmospheric Administration's Environmental Sensitivity Index dataset.




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- Natural and harder to develop
- Natural
- Partially armored and harder to develop
- Partially armored
- Armored

*Table 28: Indicator values for coastal shoreline condition within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

| | Indicator Values | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Natural and harder to develop | 30,548 | <0.1% | |
| | Natural | 61,163 | <0.1% | ↑ In good condition |
| | Partially armored and harder to develop | 1,194 | <0.1% | ↓ Not in good condition |
| | Partially armored | 4,940 | <0.1% | |
| ↓ Low | Armored | 15,683 | <0.1% | |
| | *Area not evaluated for this indicator* | *171,783,897* | *99.9%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Estuarine coastal condition

This indicator combines measures of water quality, sediment quality, contaminants in fish tissue, and benthic community condition to create an overall index of coastal estuarine condition. Estuaries serve as important nursery habitat for wildlife, including many species of fish and shellfish eaten as seafood. They also improve water quality by filtering out sediments and pollutants, provide recreational opportunities, and support coastal economies. This indicator originates from the Environmental Protection Agency's National Coastal Condition Assessment data.




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557 miles
139 279



- Good
- Fair to good
- Fair
- Poor to fair
- Poor
- Shallow estuary not assessed for condition

*Table 29: Indicator values for estuarine coastal condition within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

| | Indicator Values | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Good | 304,885 | 0.2% | |
| | Fair to good | 161,270 | <0.1% | ↑ In good condition |
| | Fair | 1,138,338 | 0.7% | ↓ Not in good condition |
| | Poor to fair | 103,359 | <0.1% | |
| | Poor | 78,889 | <0.1% | |
| ↓ Low | Shallow estuary not assessed for condition | 212,935 | 0.1% | |
| | *Area not evaluated for this indicator* | *169,897,750* | *98.8%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Gulf coral & hardbottom

This indicator predicts the presence of coral and hardbottom in the Gulf of America based on direct observations, acoustic surveys, and known locations of artificial reefs and shipwrecks. Hardbottom provides an anchor for important seafloor habitats such as deep-sea corals, plants, and sponges, providing valuable structure that supports a wide range of invertebrate and fish species. Hardbottom is also sometimes associated with diverse chemosynthetic communities supported by micro-organisms that feed off of hydrocarbon seeps. This indicator combines data from multiple sources, including Bureau of Ocean Energy Management seismic water bottom anomalies, usSEABED sediment data, several National Oceanic and Atmospheric Administration datasets, various state layers, and more.




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- Confirmed hardbottom-associated species (e.g., corals, sponges, patch reef, chemosynthetic communities)
- Confirmed natural hardbottom
- Artificial reefs
- Shipwrecks
- Probable natural hardbottom (fine resolution)
- Rock (coarse resolution)
- Gravel (coarse resolution)
- Not identified as hardbottom

*Table 30: Indicator values for Gulf coral & hardbottom within Texas. A good condition threshold is not yet defined for this indicator.*

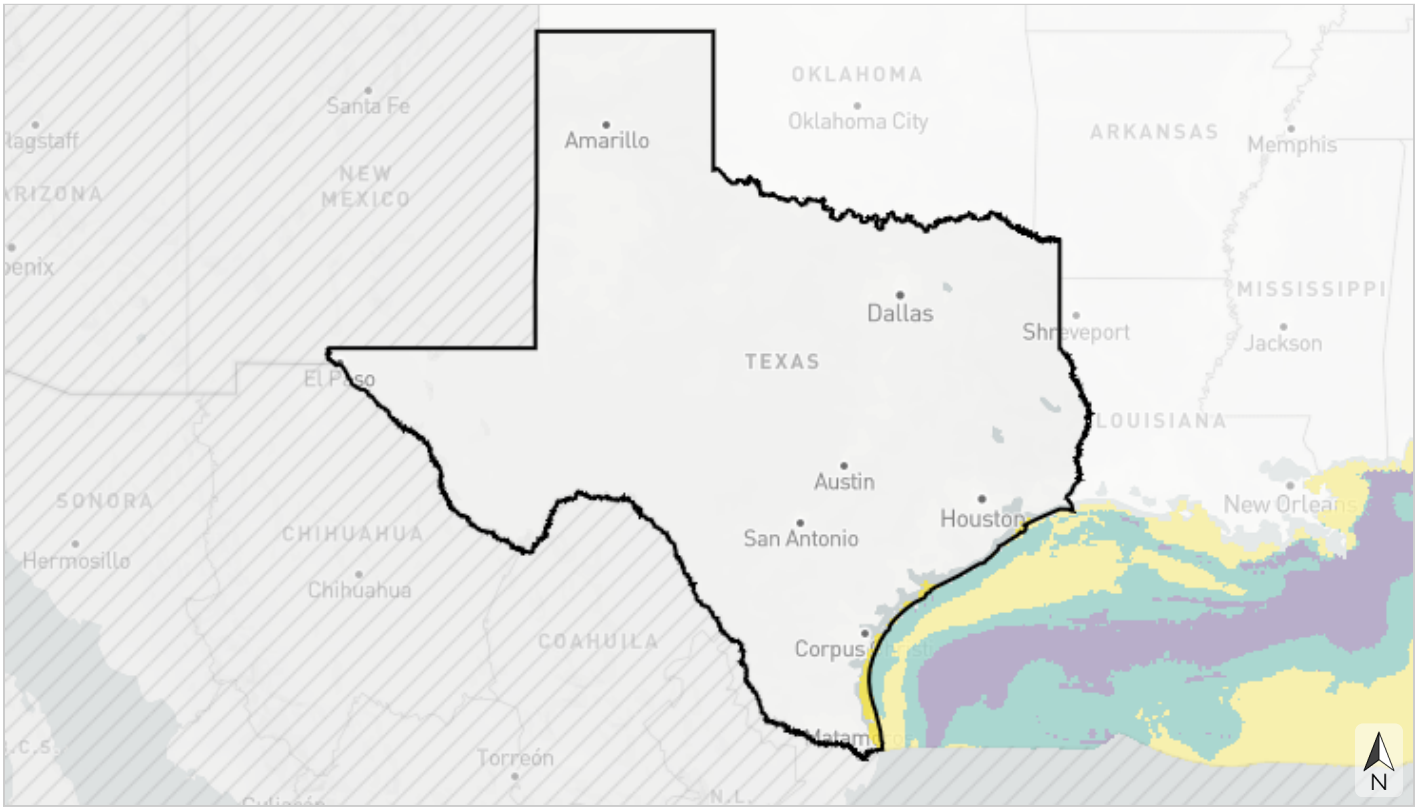
| | Indicator Values | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Confirmed hardbottom-associated species (e.g., corals, sponges, patch reef, chemosynthetic communities) | 34 | <0.1% |
| | Confirmed natural hardbottom | 0 | 0% |
| | Artificial reefs | 0 | 0% |
| | Shipwrecks | 7,557 | <0.1% |
| | Probable natural hardbottom (fine resolution) | 0 | 0% |
| | Rock (coarse resolution) | 3,189 | <0.1% |
| | Gravel (coarse resolution) | 76,304 | <0.1% |
| ↓ Low | Not identified as hardbottom | 3,418,111 | 2.0% |
| | *Area not evaluated for this indicator* | *168,392,231* | *98.0%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Gulf marine birds

This indicator identifies important areas in the Gulf of America for birds that feed exclusively or mainly at sea. It uses habitat suitability models for 24 species of marine birds (black-capped petrel, black tern, Bonaparte's gull, brown booby, brown noddy, brown pelican, bridled tern, band-rumped storm-petrel, Cory's shearwater, common loon, common tern, great shearwater, herring gull, laughing gull, masked booby, magnificent frigatebird, northern gannet, parasitic jaeger, royal tern, sandwich tern, sargasso shearwater, sooty tern, Wilson's storm-petrel, and pomarine jaeger) based on sightings from vessel-based and aerial surveys. This indicator originates from the Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS) seabird models.




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- High marine bird richness (9-16 of 24 modeled species)
- Medium marine bird richness (6-8 of 24 modeled species)
- Low marine bird richness (0-5 of 24 modeled species)

*Table 31: Indicator values for Gulf marine birds within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | High marine bird richness (9-16 of 24 modeled species) | 5,494 | <0.1% |
| | Medium marine bird richness (6-8 of 24 modeled species) | 65,508 | <0.1% |
| ↓ Low | Low marine bird richness (0-5 of 24 modeled species) | 913,765 | 0.5% |
| | *Area not evaluated for this indicator* | *170,912,659* | *99.4%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Gulf marine mammals

This indicator identifies important areas in the Gulf of America for dolphins and whales. It incorporates monthly density predictions for 13 marine mammal species or species groups (Atlantic spotted dolphin, beaked whales, blackfish [which includes killer whale, melon-headed whale, false killer whale, pygmy killer whale], bottlenose dolphin, Bryde's whale, clymene dolphin, pantropical spotted dolphin, pilot whales, pygmy/dwarf sperm whales, Rice's whale, Risso's dolphin, sperm whale, spinner dolphin) based on sightings from boat-based and aerial surveys and data on oceanographic conditions. It uses marine mammal models developed by the National Oceanic and Atmospheric Administration as part of the Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS).




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**Percentile of importance for marine mammal index species (across larger analysis area)**

- >90th percentile
- >80th-90th percentile
- >70th-80th percentile
- >60th-70th percentile
- >50th-60th percentile
- >40th-50th percentile
- >30th-40th percentile
- >20th-30th percentile
- >10th-20th percentile
- ≤10th percentile
- Land

*Table 32: Indicator values for Gulf marine mammals within Texas. A good condition threshold is not yet defined for this indicator.*

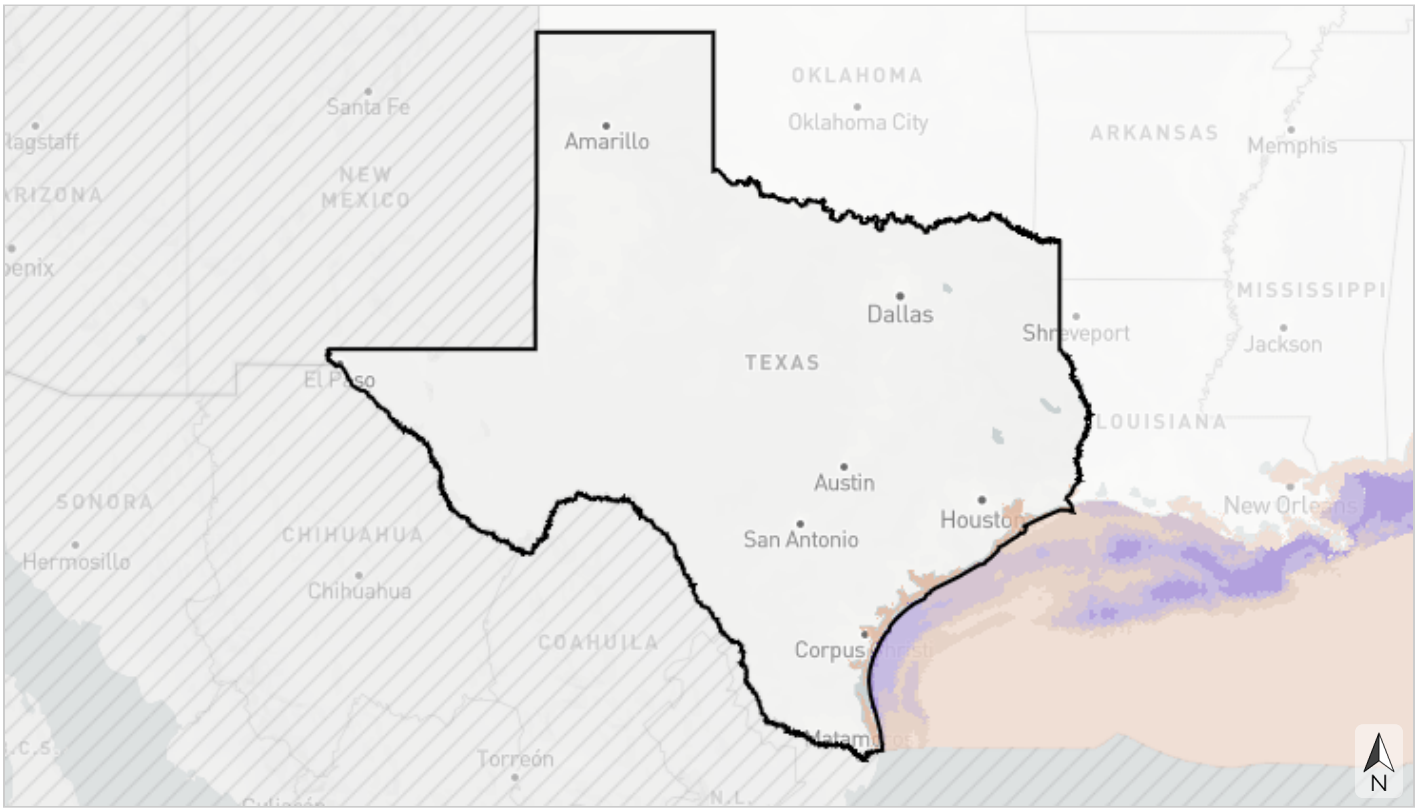
| | Indicator Values: Percentile of importance for marine mammal index species (across larger analysis area) | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | >90th percentile | 0 | 0% |
| | >80th-90th percentile | 412 | <0.1% |
| | >70th-80th percentile | 13,774 | <0.1% |
| | >60th-70th percentile | 50,207 | <0.1% |
| | >50th-60th percentile | 56,763 | <0.1% |
| | >40th-50th percentile | 178,650 | 0.1% |
| | >30th-40th percentile | 132,301 | <0.1% |
| | >20th-30th percentile | 252,137 | 0.1% |
| | >10th-20th percentile | 52,589 | <0.1% |
| | ≤10th percentile | 1,460,039 | 0.8% |
| ↓ Low | Land | 211,424 | 0.1% |
| | *Area not evaluated for this indicator* | *169,489,131* | *98.6%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Gulf sea turtles

This indicator identifies important areas in the Gulf of America for sea turtles. It incorporates monthly density predictions for four species (green, Kemp's ridley, leatherback, and loggerhead sea turtles) based on sightings from boat-based and aerial surveys and data on oceanographic conditions. It uses sea turtle models developed by the National Oceanic and Atmospheric Administration as part of the Gulf of Mexico Marine Assessment Program for Protected Species (GoMMAPPS).






**Percentile of importance for sea turtle index species (across larger analysis area)**

- >90th percentile
- >80th-90th percentile
- >70th-80th percentile
- >65th-70th percentile
- ≤65th percentile
- Land

*Table 33: Indicator values for Gulf sea turtles within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values: Percentile of importance for sea turtle index species (across larger analysis area) | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | >90th percentile | 168,995 | <0.1% |
| | >80th-90th percentile | 131,189 | <0.1% |
| | >70th-80th percentile | 209,502 | 0.1% |
| | >65th-70th percentile | 213,229 | 0.1% |
| | ≤65th percentile | 1,675,136 | 1.0% |
| ↓ Low | Land | 1,704,558 | 1.0% |
| | *Area not evaluated for this indicator* | *167,794,815* | *97.6%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Island habitat

This indicator represents important habitat for coastal island-dependent species across the Southeast. Because the isolation of islands can make them ecologically unique and protect them from disturbance and mainland predators, they often serve as important habitat for many species of mammals, plants, and insects, as well as breeding coastal birds and sea turtles. The highest scores go to island critical habitat for six threatened and endangered animal and plant species: piping plover, loggerhead sea turtle, Cape Sable thoroughwort, Florida semaphore cactus, silver rice rat, and Bartram's hairstreak butterfly. This indicator uses U.S. Fish and Wildlife Service critical habitat data and island boundaries from the U.S. Geological Survey and Esri.




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- Island critical habitat for any of six threatened and endangered species (piping plover, loggerhead sea turtle, Cape Sable thoroughwort, Florida semaphore cactus, silver rice rat, or Bartram's hairstreak butterfly)
- Other island area
- Not a coastal island

*Table 34: Indicator values for island habitat within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Island critical habitat for any of six threatened and endangered species (piping plover, loggerhead sea turtle, Cape Sable thoroughwort, Florida semaphore cactus, silver rice rat, or Bartram's hairstreak butterfly) | 84,061 | <0.1% |
| | Other island area | 270,523 | 0.2% |
| ↓ Low | Not a coastal island | 11,609,712 | 6.8% |
| | *Area not evaluated for this indicator* | *159,933,129* | *93.0%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Marine highly migratory fish

This indicator identifies important foraging and spawning areas for highly migratory fish in the Atlantic Ocean and Gulf of America. It uses physical capture and satellite tag observations, remote sensing of environmental variables, and physical oceanographic data to analyze the habitat preferences of three species (skipjack tuna, bluefin tuna, and blue shark) at various life stages. It originates from European Commission Joint Research Centre global fish models.




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**Percentile of importance for bluefin and skipjack tuna or blue shark**

- >90th percentile
- >80th-90th percentile
- >70th-80th percentile
- >60th-70th percentile
- >50th-60th percentile
- >40th-50th percentile
- >30th-40th percentile
- ≤30th percentile

*Table 35: Indicator values for marine highly migratory fish within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values: Percentile of importance for bluefin and skipjack tuna or blue shark | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | >90th percentile | 0 | 0% |
| | >80th-90th percentile | 70 | <0.1% |
| | >70th-80th percentile | 170 | <0.1% |
| | >60th-70th percentile | 1,595 | <0.1% |
| | >50th-60th percentile | 1,185 | <0.1% |
| | >40th-50th percentile | 0 | 0% |
| | >30th-40th percentile | 0 | 0% |
| ↓ Low | ≤30th percentile | 2,363 | <0.1% |
| | *Area not evaluated for this indicator* | *171,892,042* | *100.0%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Resilient coastal sites

This indicator depicts the capacity of coastal habitats to migrate to adjacent lowlands in order to sustain biodiversity and natural services under increasing inundation from sea level rise. It is based on the physical and condition characteristics of current tidal complexes, their predicted migration space, and surrounding buffer areas. These characteristics include marsh complex size, shared edge with migration space, sediment balance, water quality, natural landcover, landform diversity, and more. This indicator originates from The Nature Conservancy's Resilient Coastal Sites project.




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- Most resilient
- More resilient
- Slightly more resilient
- Average/median resilience
- Slightly less resilient
- Less resilient
- Least resilient
- Sea level rise area

*Table 36: Indicator values for resilient coastal sites within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Most resilient | 54,537 | <0.1% |
| | More resilient | 1,337,864 | 0.8% |
| | Slightly more resilient | 417,415 | 0.2% |
| | Average/median resilience | 60,210 | <0.1% |
| | Slightly less resilient | 2,543 | <0.1% |
| | Less resilient | 819 | <0.1% |
| | Least resilient | 1,958 | <0.1% |
| ↓ Low | Sea level rise area | 141,223 | <0.1% |
| | *Area not evaluated for this indicator* | *169,880,857* | *98.8%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Seagrass

This indicator represents the presence of seagrass in the Atlantic Ocean and Gulf of America. Seagrasses provide food and habitat for a range of marine and estuarine wildlife, including fish, sea turtles, shrimp, crabs, oysters, and more. They also produce oxygen, filter water, control erosion, and buffer storms. Seagrasses serve as an important indicator of the overall health of coastal ecosystems because they are sensitive to water quality and require sufficiently clear water for sunlight to penetrate. This indicator originates from the National Oceanic and Atmospheric Administration's Marine Cadastre.




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Seagrass present

*Table 37: Indicator values for seagrass within Texas. A good condition threshold is not yet defined for this indicator.*

| | Indicator Values | Acres | Percent of Area |
|---|---|---|---|
| ↑ High | Seagrass present | 228,430 | 0.1% |
| | *Area not evaluated for this indicator* | *171,668,995* | *99.9%* |
| | **Total area** | **171,897,425** | **100%** |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

Coastal & marine

# Stable coastal wetlands

This indicator uses remote sensing to calculate the unvegetated-vegetated ratio of tidal wetlands, which compares how much of a wetland is not covered by plants (e.g., sediment, rocks, open water) to how much is covered by plants. Marshes that maintain a higher proportion of vegetation tend to be more stable and resilient to threats like sea level rise, erosion, and coastal development. This ratio, and how it changes over time, is a good surrogate for salt marsh degradation processes like sediment loss and conversion to open water. This indicator originates from a U.S. Geological Survey project on an unvegetated to vegetated ratio for coastal wetlands.




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- Stable coastal wetlands
- Other coastal wetlands
- Not identified as coastal wetlands

*Table 38: Indicator values for stable coastal wetlands within Texas. Good condition thresholds reflect the range of indicator values that occur in healthy, functioning ecosystems.*

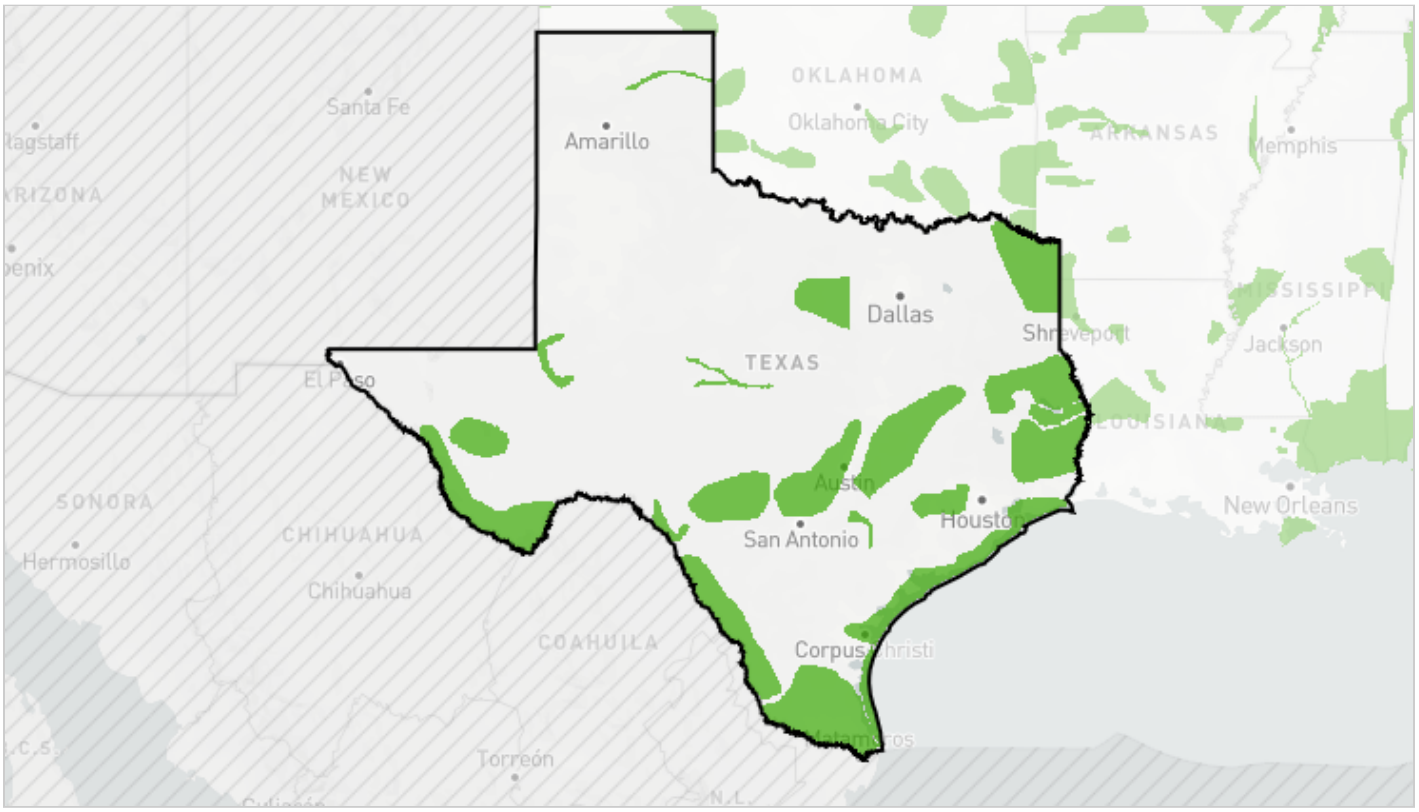
| | Indicator Values | Acres | Percent of Area | |
|---|---|---|---|---|
| ↑ High | Stable coastal wetlands | 302,417 | 0.2% | ↑ In good condition |
| | Other coastal wetlands | 390,136 | 0.2% | ↓ Not in good condition |
| ↓ Low | Not identified as coastal wetlands | 2,441,195 | 1.4% | |
| | *Area not evaluated for this indicator* | *168,763,677* | *98.2%* | |
| | **Total area** | **171,897,425** | **100%** | |

To learn more and explore the GIS data, view this indicator in the SECAS Atlas.

# More Information

## Priority Amphibian and Reptile Conservation Areas

Priority Amphibian and Reptile Conservation Areas (PARCAs) are an expert-driven, nonregulatory designation that includes places capable of supporting viable amphibian and reptile populations, places occupied by rare or imperiled species, and places rich in biodiversity or species unique to that geographic area. Reptiles and amphibians are a critical part of the Southeast region's rich biodiversity and many populations are declining in the face of threats like habitat loss, invasive species, and climate change. The PARCA dataset is maintained by the [Amphibian and Reptile Conservancy](Amphibian and Reptile Conservancy) and does not yet include Virginia or Kentucky.




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Priority Amphibian and Reptile Conservation Area (PARCA)

Not a PARCA (excluding Kentucky and Virginia)

*Table 39: Extent of Priority Amphibian and Reptile Conservation Areas within Texas.*

| Priority Amphibian and Reptile Conservation Areas status | Acres | Percent of Area |
|---|---|---|
| Priority Amphibian and Reptile Conservation Area (PARCA) | 34,692,370 | 20.2% |
| Not a PARCA (excluding Kentucky and Virginia) | 137,186,329 | 79.8% |
| **Total area** | **171,897,425** | **100%** |

## Priority Amphibian and Reptile Conservation Areas at this location:

**Alta Tamaulipas**

The Lower Rio Grande Valley contains mostly thorn scrub and live oak woodlands, with grasslands mixed in mostly on deep, sandy soil. Scattered riparian woodlands occur along the Rio Grande and near oxbow lakes. This PARCA represents a convergence between North and South America, with many unique Central American species like the northern cat-eyed snake, speckled racer, Mexican burrowing toad, and the sheep frog occurring here at the northernmost extents of their ranges. Threats include increased mortality and dispersal limitations from border wall construction, urbanization, increased agricultural development, and climate change-worsened drought.

**Balconian**

Like many areas of Texas, Balconia has an impressive collection of herpetofaunal diversity, but this PARCA is a top national priority, as it also includes a massive diversity of cave salamanders that are still being described. These cave specialists are microendemic and are all facing shared threats due to the water table being lowered and local pollution, to which they are highly sensitive. Other terrestrial species are at risk from invasive species (fire ants, exotic grasses, feral cats) and nearby urbanization, which continues to convert and fragment essential habitats. Wildlife species, above and below ground, will benefit from the improved hydrology that will be a focus of conservation efforts in this PARCA.

**Beaver Grasslands**

This PARCA includes part of the Shortgrass Prairie Region, which is home to a diverse range of herbaceous species. Much shortgrass prairie has been converted to other land uses, such as crop fields, ranching, housing developments, and roads. Working together with partners and landowners to convert old crop fields back to shortgrass prairies using native plants and removing exotic invasive species like Russian thistle and old world bluestem is a necessary step to mending this fragmented ecosystem.

**Big Bend Corridor**

A famous hotspot of biodiversity, the Big Bend Corridor boasts a rich assortment of herpetofauna. Species such as the Texas horned lizard, barred tiger salamander, and Mojave rattlesnake populate diverse habitats, including Chihuahuan Desert scrub, desert grasslands, and Madrean evergreen woodlands. With an average of less than 16 inches of rain annually over much of this PARCA, amphibian and reptile distributions are heavily influenced by available moisture. Surface water rights, aquifer drawdowns, and spring-fed water source alterations are a threat to herpetofauna, alongside increased vehicular and construction traffic along the border. Additional threats impacting riparian and wetland areas include invasive plants and free-ranging livestock populations. Though this PARCA contains large tracts of public land, such as Big Bend National Park, much of it is in private holdings, and conservation efforts require private engagement.

**Big Thicket**

Covering the southern Pineywoods, the Big Thicket PARCA is primarily forested but contains a diversity of habitats and variation with some open habitats, such as prairies and sandstone barrens. Vegetation communities today are very different from what was here pre-settlement, and historically mesophytic forests with defined mid and understories occurred on steep river bluffs. These forests featured large hardwoods, such as American beech, southern magnolia, and white oak. Modern management for timber

production has shifted these forests to be generally younger and more dominated by loblolly pine. Xeric sandhills occur on deep riparian sand deposits and feature dry adapted plants like cacti and yuccas. The southernmost extent of longleaf pine in Texas extends into this PARCA. Herpetofauna in this PARCA is predominantly eastern, with species like the spotted dusky salamander, pig frog, and northern scarlet snake reaching the westernmost extents of their ranges. Threats include unsustainable timber practices, fire suppression, invasive species, roadway expansions, and trotlining (stringing up fishing lines with hooks at regular intervals).

**Caddoan**
The Caddoan PARCA in northeast Texas represents a variety of habitats; the eastern portion is generally forested with shortleaf pine, oaks and hickories, transitioning into post oak savanna and prairie in the western portions. Historically, the prairies contained a diverse microtopography of "mima" mounds and depressions. Much of this has been lost as they were converted to agriculture, but some examples on conservation and managed private lands remain. River floodplains contain some of the best bottomland hardwood forest habitat in the state of Texas. Species such as the pygmy rattlesnake, crawfish frog, and the Gulf Coast waterdog call this PARCA home, and they’re threatened by fire suppression, forest conversion and clearing, prairie succession, mining, and invasive species.

**Coastal Prairie**
With mid to tallgrass prairie over a gently rolling topography and ephemeral prairie "pothole" wetlands, the Coastal Prairie PARCA contains some of the best remaining habitat for species such as the crawfish frog, slender glass lizard, and historically, the Houston toad. Rivers and streams provide habitat for species like the alligator snapping turtle and Texas map turtle, and drainages east of the Brazos river are important for flood control in the Houston metropolitan area. Threats in this PARCA include urbanization, non-native "improved" grasses, fire suppression, unsustainable agricultural practices, invasive species, and altered hydrology.

**Concho**
The Concho PARCA contains the Concho and Colorado River basins in central Texas as well as their riparian corridors. This PARCA contains the range of the Concho water snake, a Texas endemic species found nowhere else but these river drainages. Natural riffles with lots of rocks and crevices are critical for maintaining populations of minnows, aquatic insects, and crayfish, important prey species for aquatic herpetofauna. Maintaining water quality and habitat integrity is crucial for the biodiversity of the Concho PARCA, as well as for the human populations that depend on its water.

**Davis Mountains**
Rising from the surrounding deserts of West Texas, the Davis Mountains provide a "sky island," with cooler temperatures and increased precipitation at higher elevations allowing forests and other mesic habitats to persist. Largely dominated by montane grassland, woodlands typically become more prevalent at higher elevations. A large, unique assemblage of herpetofauna make their home here, including species such as the gray-banded kingsnake, reticulate banded gecko, and canyon tree frog. The vast majority of land in this PARCA is private, and threats include increasingly dense development in the region. Additional threats include altered fire regimes and climate change.

**Edwards Eurycea**
The Edwards Eurycea PARCA sits atop the Edwards Aquifer, a vast groundwater resource that has shaped the development of south-central Texas and today serves nearly two million users. More than 40 different subterranean adapted species occur here across many different animal groups, including catfish, amphipods, snails, and of course, salamanders. Many groundwater salamanders occur here, and their relationship to each other is still being worked out. Unique species like the Texas blind salamander are among the most fascinating inhabitants. The chief threats to the Edwards Aquifer and its biodiversity are reduced spring flows caused by increased pumping and water quality degradation from increased development.

**Guadalupe San Marcos**
The Guadalupe San Marcos PARCA follows the rivers of its name from south of Luling, through their confluence, downstream south of Cuero, TX. These rivers are some of Texas' most popular recreation areas, and they’re also a hotspot of aquatic biodiversity. Some threats to this PARCA include overuse for recreation (especially in sensitive areas), water use, pollution, and water quality degradation. These challenges must be addressed to ensure the river remains safe and healthy for biodiversity as well as human use.

**Laredo Reach**
The Laredo Reach PARCA is mostly contained in the Tamaulipan Thornscrub ecoregion, a very arid landscape largely lacking surface waters other than the Rio Grande. Sparse to degraded grasslands may be found occasionally, with riparian vegetation, both native and exotic, along the banks of the Rio Grande. Terrestrial species include the Berlandier's tortoise, Texas indigo snake, and reticulate collared lizard, while aquatic species like the Rio Grande cooter utilize the river and riparian habitats. Threats to the biodiversity of Laredo Reach include converting thornscrub to improved pasture, increased development, fencing obstacles (particularly to tortoises), and invasive riparian vegetation like giant cane.

**Longleaf Ridge**
The Longleaf Ridge PARCA is located entirely within the Pineywoods, and comprises the heart of the range of longleaf pine in Texas. It also contains the majority of the known range of the Louisiana pine snake within the state. Portions of Angelina and Sabine National Forests are contained within this PARCA, as well as a large area of conservation easements. The northern half of this PARCA contains longleaf pine savannas on sandy, rolling uplands, containing species such as northern scarlet snakes, while shifting flatter towards the south. Downslope from longleaf savannas, hardwood forests, bald cypress-dominated floodplains, and seepage slopes occur, providing home for species such as the alligator snapping turtle, spotted dusky salamander, and pickerel frog. Threats to the biodiversity of this PARCA include unsustainable timber practices, forest conversion and clearing, invasive species, off-road vehicle use, and fire suppression.

**Lower Little River**
This important area for reptile and amphibian diversity has an intricate system of large streams, oxbow lakes, and backswamps. Covering hilly dissected uplands, floodplains, and low terraces are various species of oak, pine and cypress trees. This PARCA includes Little River National Wildlife Refuge, which is home to 11 state Champion Trees (largest of their respective species in the state) and offers refuge to

several at-risk turtle species.

**Monahans Sandhills**
The Monahans Sandhills PARCA follows a unique formation, created thousands of years ago when sandy floodplains dried up and winds accumulated them in large sand dunes. Today, many of the dunes are vegetated with grasses and shinnery oak, which provides structure, shade, food, and breeding grounds for species like the dunes sagebrush lizard in the harsh landscape. A chief threat to this PARCA is the rapid boom in oil and gas development. With the 2024 listing of the dunes sagebrush lizard as federally endangered under the Endangered Species Act, cooperation with private industry will be crucial to preserve the natural resources of the Monahans Sandhills PARCA.

**No Man's Land**
The No Man’s Land PARCA contains unique geologic formations occurring in northeast to southwest bands across the area. The Jackson, Catahoula, Cook Mountain, and Fleming formations present distinctive soil types and conditions that influenced the development of natural community types along these formation bands. Calcareous clays, sandstones, saline deposits, siltstones, and ironstones have shaped the development of natural communities such as the calcareous forests, calcareous prairies, saline prairies, and small stream forests of this area. The south and southwestern portions of this PARCA are known for western longleaf pine flatwoods savannas and associated flatwoods ponds, and this area serves as the transition zone between Louisiana’s coastal prairies and upland longleaf pine woodlands. Threats to this PARCA include fire suppression, invasive species, siltation, and hydrology changes to ephemeral ponds. Major priorities for this PARCA include opening pine forests to establish natural canopy densities, returning natural burn regimes, maintaining best management practices and streamside management zones, creating/restoring ephemeral ponds, and controlling invasive species.

**Padre Island**
This dynamic barrier island system, largely managed by the National Park Service as Padre Island National Seashore, is located on the longest section of undeveloped barrier island in the United States. This system protects the majority of the remaining Texas coastal prairie, a dynamic environment constantly sculpted by wind and sea, and the Laguna Madre, one of the few hypersaline lagoon environments left in the world. More than 60% of this area consists of wetlands, including marshes, inland waters, wind tidal flats, and seagrass beds. These beaches are highly critical nesting habitat for many federally endangered sea turtle species such as the loggerhead, leatherback, green, hawksbill, and Kemp's ridley. This area is also home to many important terrestrial species like the Texas indigo snake and the black spotted newt. Controlling invasive species, reducing off-road vehicles, and limiting poaching will help reduce nest disturbance and destruction, but climatic threats such as sea-level rise, algal blooms, increasing storm frequency, and high sand temperatures must be addressed through coastal forest remediation and management of human activities.

**Pronatura**
The Pronatura PARCA is bounded by perennial surface water drainages connected to the Rio Grande Basin. At the intersection of three ecoregions, this PARCA contains influences from the Tamaulipan Thornscrub, Edwards Plateau, and Chihuahuan Desert. For this region's unique species, such as endemic groundwater salamanders and aquatic turtles like the Rio Grande cooter, water is key. They face threats

such as water overextraction, water quality degradation, and invasive species. Terrestrial species, such as the Berlandier's tortoise and Texas indigo snake, face threats from historical land use changes, land clearing, and range improvement practices. Most of this PARCA is comprised of private ranchland, making cooperation with landowners critical to maintaining its biodiversity.

**Sandy Sanders**
This unique PARCA is a rolling grassland located in a less populated area of Oklahoma and in one of the warmest and driest parts of the state. Most of the land in this region is privately owned ranchland, except for a 30,000-acre state Wildlife Management Area. Mixed-grass prairie and prairie shrublands with an overstory of honey mesquite are found throughout this region. Redberry juniper woodlands and gypsum canyonland habitats with rare plant species are found on hills and canyons peppered throughout the area. Fire suppression and poor range management have encouraged brush encroachment within the prairie, leading to habitat degradation. Rattlesnake roundups threaten local populations of rattlesnakes such as the western diamondback. These top predators play a major role in their ecosystems and already face many other threats, including habitat loss, road mortality, and poaching.

**Shinnery Oaks**
The Shinnery Oaks PARCA is a unique landscape comprised of eroded canyonlands, old sand dunes dominated by shinnery oak, mixed-grass prairies, and riparian woodlands along perennial and intermittent streams. Historically, these shrublands were once estimated to cover about 750,000 acres of Oklahoma, but now only 116,000 acres remain. The mixed-grass prairies in this area are formed by a diverse plant composition, but little bluestem and sideoats grama are the dominant grass species. Most of this important habitat has been converted to crops and ranchland, and heavy grazing coupled with fire suppression has led to an invasion of introduced grasses and forbs. Fragmentation caused by roads is another common issue in this area, and wildlife road mortalities can be detrimental to struggling species. Figuring out a solution to minimizing road fatalities, especially with hard-to-see reptiles and amphibians, will be an objective for this PARCA.

**Stink Finger**
The Stink Finger PARCA primarily consisted of shortleaf pine-oak-hickory woodland until post-settlement, when the majority of this community type was removed, and it has since been converted to loblolly pine plantation. Some natural stands of shortleaf pine-oak-hickory woodland still exist in this ecoregion. The major habitat types found within this PARCA consist of batture forests, hardwood slope forests, mixed hardwood-loblolly pine forests, shortleaf pine-oak-hickory woodlands, xeric sandhill woodlands, hardwood flatwoods, calcareous forests, calcareous prairies, saline prairies, and sandbars. The Shreveport Airport contains the last known population of southern crawfish frogs in the state. Threats to this PARCA include urbanization, vehicle mortality, invasive species, and habitat conversion. Major priorities for this PARCA include restoring native habitat types, controlling invasive species, and creating habitat for the southern crawfish frog.

**Sugar Sands**
The Sugar Sands PARCA follows a band of deep, sandy soils that runs from the Colorado River northeast to the Trinity River. Defined at its southern boundary by the Lost Pines, a relict population of loblolly pine that is now isolated from populations further east, this area gradually transitions into post oak savannas,

which along with a very high diversity of reptiles and amphibians, are rapidly disappearing. This PARCA covers what is generally thought to be the range of the federally endangered Houston toad in the northwestern part of its distribution. Some of the major threats here include fire suppression, invasive species (feral hogs), overgrazing, and human development. By working with universities and land managers, we hope to better understand how to mitigate the disappearance of critical wildlife habitats such as the post oak savanna.

**Sulfur River**
The Sulfur River PARCA is composed of rolling plains that are broken by nearly flat fluvial terraces, bottomlands, sandy low hills, and low cuestas. One of the biggest bottomland hardwood habitats persisting along the Red River Valley is found within this PARCA and is home to a thriving alligator population. Recreational activities are abundant in this region, but maintaining the water quality and protecting important wetlands will help defend the abundant wildlife.

**Texas Bays and Marshes**
The Texas Bays and Marshes PARCA follows the coastline from the border with Louisiana all the way southwest to the Corpus Christi Bay. An extensive assortment of habitats, including fresh and saltwater marshes, hardwood bottomlands, and oak woodlands, provide habitat for many species, including the diamondback terrapin, Texas indigo snake, and black spotted newt. Some proximate human-caused threats here include coastal development, fire suppression, freshwater inflow to estuaries, and barge traffic. However, climatic threats such as sea-level rise and increased frequency and severity of tropical storm systems must also be addressed.

**Texas Canadian River**
True to its name, the Canadian River PARCA contains the Canadian River from below the Lake Meredith Dam all the way to the Oklahoma border, as well as adjacent riparian zone and floodplain oxbow lakes. The riparian forests are dominated largely by cottonwoods and hackberries with some smaller tributaries containing mesquite-juniper brush. Beyond the riparian zone, there are large bands of mesquite shrub-grassland. Species such as the smooth softshell, Texas horned lizard, and plains hognose snake occupy this PARCA and are threatened by invasive plants and increased agricultural and urban water uses.

**Texas Pineywoods**
This large PARCA includes the central Pineywoods and contains the majority of Davy Crockett, Angelina, and Sabine National Forests. A rich diversity of habitats occur here, from longleaf pine savanna uplands to hardwood stratified slopes, deep coarse sandhills, and bald cypress-tupelo bottomlands. Much of these historical habitats have been converted and lost, but large patches remain on national forest land and private conservation easements. Species such as the pygmy rattlesnake, mole salamander, and western chicken turtle represent the area's rich biodiversity, and are threatened by unsustainable timber practices, fire suppression, invasive plants and feral hogs, off-road vehicle use, and loss of prairie remnants.

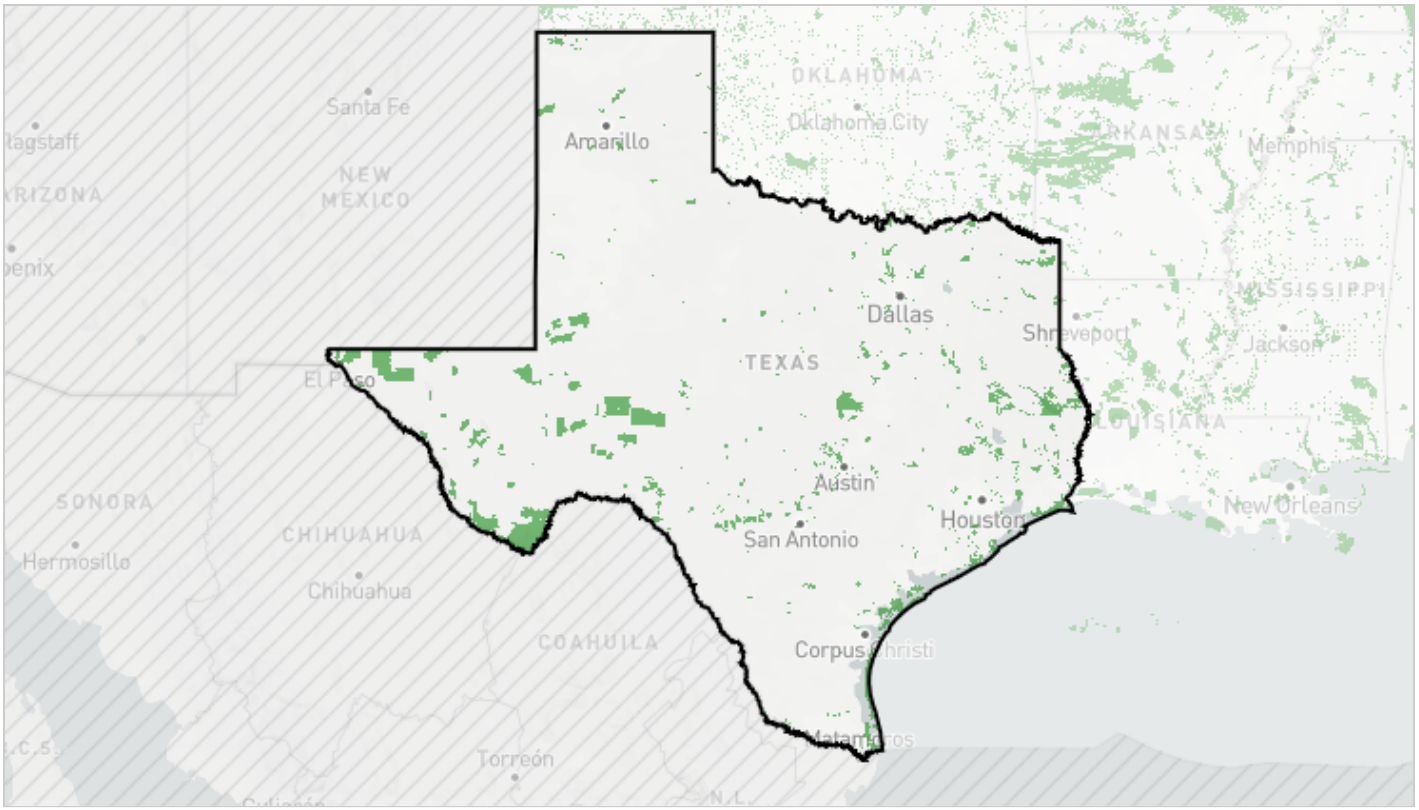
**Western Cross Timbers and Prairies**
Located west of the Dallas/Fort Worth metro area, this PARCA contains a mix of tall to midgrass prairie remnants, western cross timbers, and riverbottom forests. This region has undergone extensive changes in the past 150 years, with timbered areas representing significant barriers to settlers traveling the open prairies. Much of the range of the Brazos water snake is contained within this PARCA, solely in the upper

Brazos River drainage. Alongside this Texas endemic are species such as the eastern massasauga, Woodhouse's toad, and ornate box turtle. Threats to this PARCA include urbanization, fire suppression, oil and gas development, invasive species, and altered hydrologic regimes.

Names and descriptions of public Priority Amphibian and Reptile Areas were provided by the Amphibian and Reptile Conservancy on August 30, 2024 and edited slightly for clarity and consistency.

# Protected Areas

Protected areas include a variety of public lands owned or managed by federal, state, and local agencies and nonprofit organizations, as well as some private conservation lands and conservation easements. Protected areas are derived from the Protected Areas Database of the United States (PAD-US v4.1) and include Fee, Designation, Easement, Marine, and Proclamation (Dept. of Defense lands only) boundaries.




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Within a protected area

Not within a protected area

*Table 40: Extent of protected areas within Texas.*

| Protected area status | Acres | Percent of Area |
|---|---|---|
| Not within a protected area | 162,875,933 | 94.8% |
| Within a protected area | 9,021,492 | 5.2% |
| **Total area** | **171,897,425** | **100%** |

## Protected areas at this location:

- Permanent University Fund (Texas General Land Office; 2,186,002 acres)
- Big Bend National Park (National Park Service; 784,238 acres)
- Big Bend Proposed or Recommended Wilderness Area (576,637 acres)
- The Nature Conservancy Conservation Easement (Private Land Owner; 403,736 acres)
- Big Bend Ranch State Park (Texas Parks and Wildlife Department; 313,672 acres)
- Big Bend Ranch State Park (State; 313,333 acres)
- Fort Cavazos (219,324 acres)
- Mission-Aransas National Estuarine Research Reserve (185,674 acres)
- Sam Houston National Forest (USDA Forest Service; 163,247 acres)
- Sabine National Forest (USDA Forest Service; 161,158 acres)
- Davy Crockett National Forest (USDA Forest Service; 161,094 acres)
- Angelina National Forest (USDA Forest Service; 154,309 acres)
- Padre Island National Seashore (131,161 acres)
- Padre Island National Seashore (National Park Service; 130,489 acres)
- Fort Bliss (123,240 acres)
- Sam Rayburn Recreation Area (116,718 acres)
- Aransas National Wildlife Refuge (116,538 acres)
- Sam Rayburn Reservoir (113,190 acres)
- Big Thicket National Preserve (National Park Service; 108,722 acres)
- Black Gap Wildlife Management Area (Texas Parks and Wildlife Department; 104,641 acres)
- Black Gap Wildlife Management Area (State; 104,637 acres)
- Laguna Atascosa National Wildlife Refuge (US Fish and Wildlife Service; 100,401 acres)
- Lower Rio Grande Valley National Wildlife Refuge (98,069 acres)
- Laguna Atascosa National Wildlife Refuge (91,401 acres)
- Aransas National Wildlife Refuge (US Fish and Wildlife Service; 90,788 acres)
- ... and 18,818 more protected areas ...

Note: areas are listed based on name, ownership, and boundary information in the Protected Areas Database of the United States, which may include overlapping and duplicate areas. Only protected areas with at least 1 acre of overlap with this area are included in the list above.

## Sea Level Rise

NOAA's sea level rise (SLR) inundation models represent areas likely to experience flooding at high tide based on each foot of SLR above current levels. Darker blue areas will experience flooding first, and at greater depth, compared to lighter blue areas. These models are not linked to a future timeframe; see the projections below. NOAA calculates the inundation footprint at "mean higher high water", or the average highest daily tide. The area covered in each SLR scenario includes areas projected to be inundated at lower levels. For example, the area inundated by 4 ft of SLR also includes areas inundated by 3 ft, 2 ft, 1 ft, and 0 ft of SLR (where 0 ft represents current levels).

To explore additional SLR information, please see NOAA's Sea Level Rise Viewer.




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**Flooding extent by projected sea level rise (ft)**

0 1 2 3 4 5 6 7 8 9 10

*Table 41: Extent of flooding by projected average highest daily tide due to sea level rise within Texas. Values from the NOAA sea level rise inundation data.*

| Feet of sea level rise | Acres | Percent of Area |
|---|---|---|
| 0 feet | 2,478,465 | 1.4% |
| 1 foot | 2,809,554 | 1.6% |
| 2 feet | 3,107,056 | 1.8% |
| 3 feet | 3,316,412 | 1.9% |
| 4 feet | 3,483,160 | 2.0% |
| 5 feet | 3,630,119 | 2.1% |
| 6 feet | 3,782,377 | 2.2% |
| 7 feet | 3,950,305 | 2.3% |
| 8 feet | 4,111,739 | 2.4% |
| 9 feet | 4,273,813 | 2.5% |
| 10 feet | 4,433,222 | 2.6% |
| *Not projected to be inundated by up to 10 feet* | *10,499,022* | *6.1%* |
| *Sea level rise is unlikely to be a threat (inland counties)* | *156,958,840* | *91.3%* |
| *No sea level rise data available* | *6,340* | *<0.1%* |
| **Total area** | **171,897,425** | **100%** |

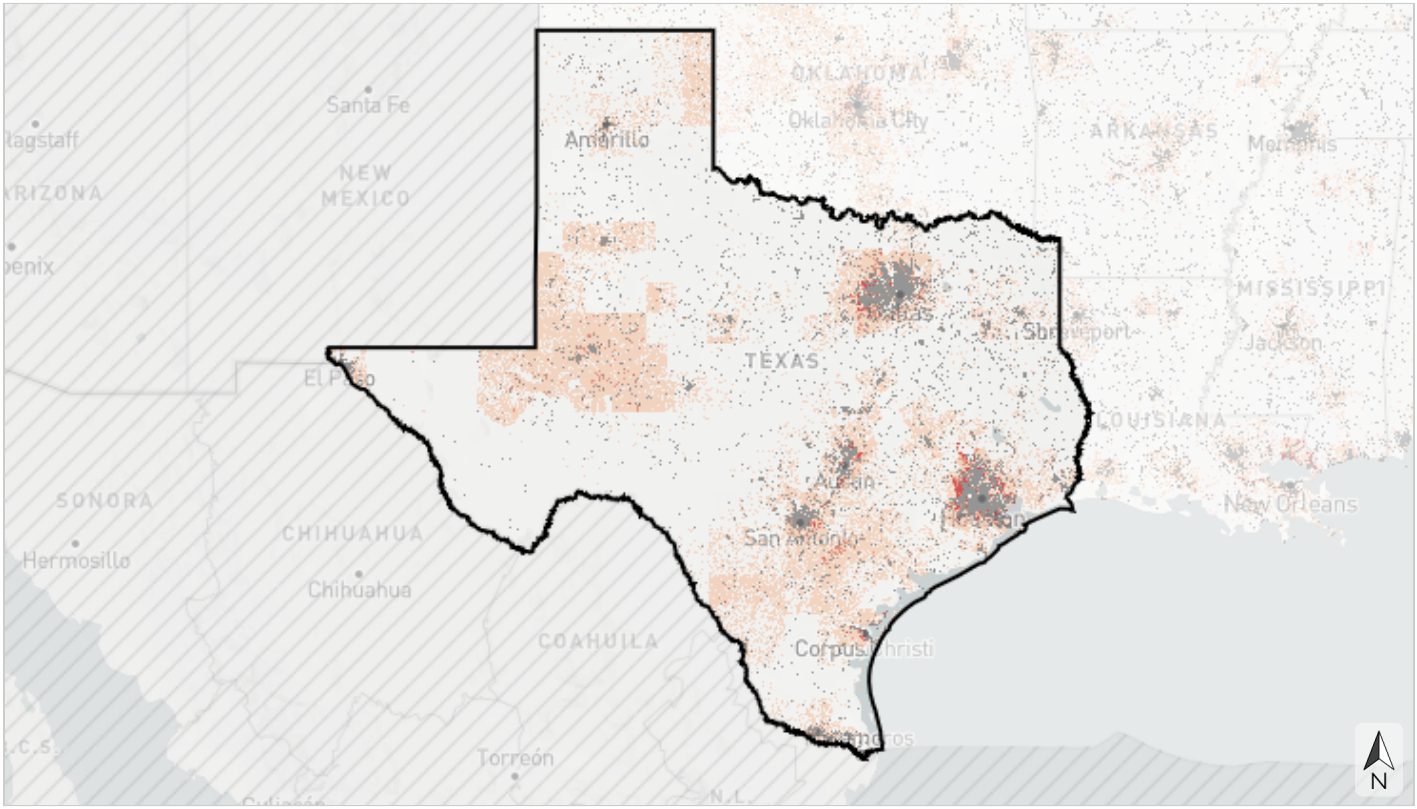
*Table 42: Projected sea level rise by decade within Texas. Values are based on area-weighted averages of decadal projections for 1-degree grid cells that overlap this area based on NOAA's 2022 Sea Level Rise Report. 2060 corresponds to the SECAS goal: a 10% or greater improvement in the health, function, and connectivity of Southeastern ecosystems by 2060.*

| SLR Scenario | 2020 (ft) | 2030 (ft) | 2040 (ft) | 2050 (ft) | 2060 (ft) | 2070 (ft) | 2080 (ft) | 2090 (ft) | 2100 (ft) |
|---|---|---|---|---|---|---|---|---|---|
| Low | 0.5 | 0.76 | 1 | 1.3 | 1.5 | 1.7 | 1.9 | 2.1 | 2.3 |
| Intermediate-low | 0.53 | 0.82 | 1.1 | 1.4 | 1.7 | 2 | 2.3 | 2.6 | 2.9 |
| Intermediate | 0.54 | 0.85 | 1.2 | 1.5 | 1.9 | 2.4 | 3 | 3.6 | 4.4 |
| Intermediate-high | 0.55 | 0.87 | 1.3 | 1.7 | 2.4 | 3.1 | 4 | 5 | 6 |
| High | 0.55 | 0.9 | 1.4 | 1.9 | 2.8 | 3.8 | 5 | 6.3 | 7.7 |

## Urban Growth

The FUTURES urban growth model predicts the likelihood that an area will urbanize at every decade from 2020 to 2100. Developed areas from the 2021 National Landcover Database serve as the baseline for current urban areas. The model simulates landscape change based on trends in population growth, local development suitability factors, and an urban patch-growing algorithm. It considers environmental drivers like distance to floodplain, slope, and available infrastructure, and even socio-economic status. The probability of urbanization for each area reflects how many times it urbanized out of 50 model runs.

To explore maps for additional time periods, click here.




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**Probability of urbanization by 2060**

- Urban in 2021
- Very high likelihood of urbanization (>50% probability)
- High likelihood of urbanization (25 - 50% probability)
- Moderate likelihood of urbanization (2 - 25% probability)
- Not likely to urbanize

*Table 43: Extent of projected urbanization by decade within Texas. Values from FUTURES model projections for the contiguous United States developed by the Center for Geospatial Analytics, NC State University. 2060 corresponds to the SECAS goal: a 10% or greater improvement in the health, function, and connectivity of Southeastern ecosystems by 2060.*

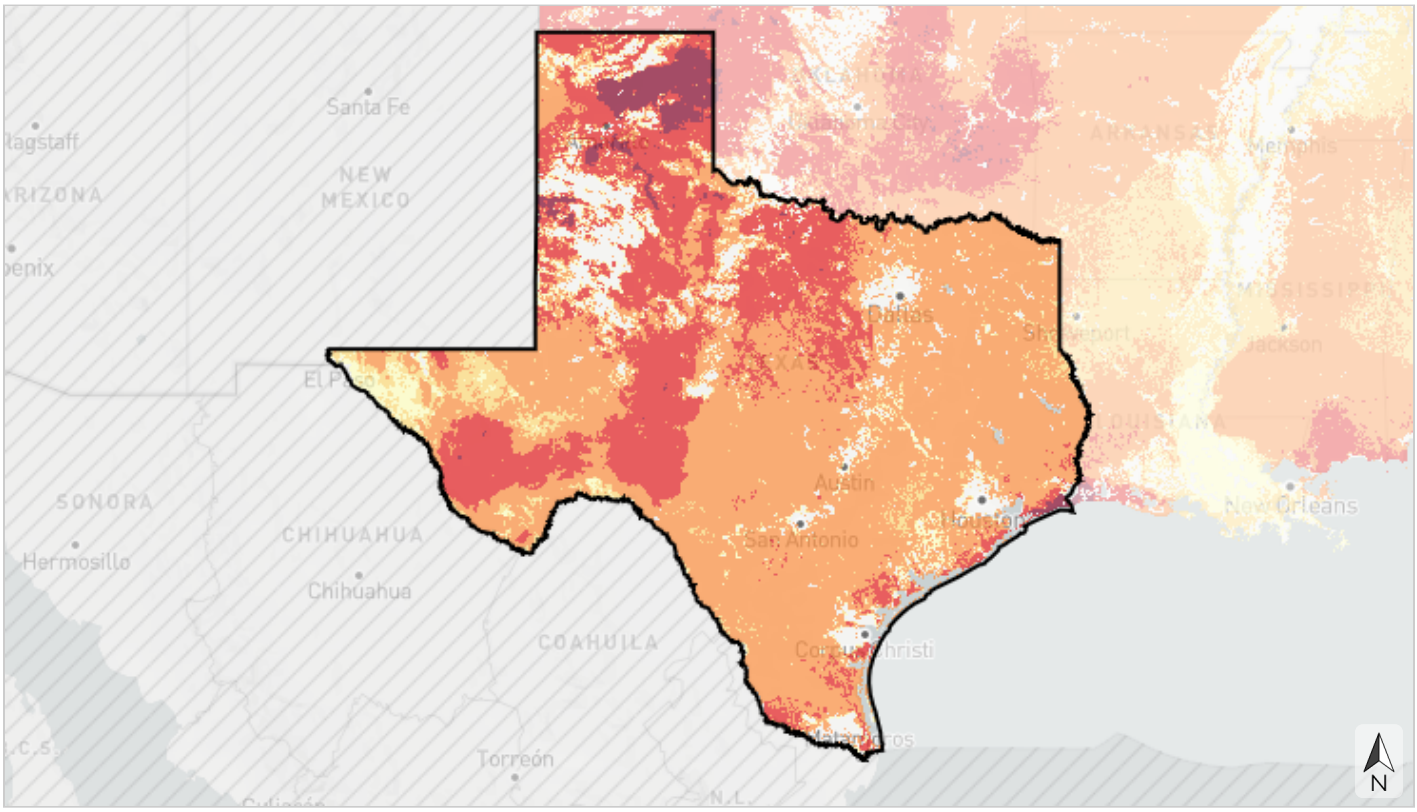
| Decade | Acres | Percent of Area |
|---|---|---|
| Urban in 2021 | 10,284,773 | 6.0% |
| 2030 projected extent | 11,155,182 | 6.5% |
| 2040 projected extent | 11,723,021 | 6.8% |
| 2050 projected extent | 12,219,062 | 7.1% |
| 2060 projected extent | 12,671,220 | 7.4% |
| 2070 projected extent | 13,045,617 | 7.6% |
| 2080 projected extent | 13,333,696 | 7.8% |
| 2090 projected extent | 13,535,923 | 7.9% |
| 2100 projected extent | 13,673,739 | 8.0% |
| *Not projected to urbanize by 2100* | *158,204,960* | *92.0%* |
| *No urbanization data available* | *18,726* | *<0.1%* |
| **Total area** | **171,897,425** | **100%** |

6.0% of this area is already urban in 2021, and an additional 19.5% has at least a moderate probability of urbanizing by 2060.

By 2060, the size of the urban footprint is projected to increase **23%** over 2021 levels.

# Wildfire Likelihood

Wildfire likelihood data originate from the Wildfire Risk to Communities project developed by the U.S. Forest Service. This layer depicts the probability of wildfire burning in a specific location in any given year. Annual burn probabilities in the United States range from 0-14%, but do not exceed 8% in the Southeast. Wildfire likelihood is based on fire behavior modeling across thousands of simulations of possible fire seasons. In each simulation, factors contributing to the probability of a fire occurring (such as weather, topography, and ignitions) vary based on patterns derived from observations in recent decades. Wildfire likelihood is not predictive and does not reflect any forecasted future weather or fire danger conditions. It also does not say anything about the intensity of fire if it occurs. To explore additional wildfire risk information, please see the Wildfire Risk to Communities website.




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**Wildfire likelihood (annual burn probability)**

- High
- Moderate-high
- Moderate
- Low-moderate
- Low
- Not predicted to experience wildfire

*Table 44: Area in each wildfire probability category within Texas. Values from the Wildire Risk To Communities project developed by the USDA Forest Service.*

| Wildfire likelihood (annual burn probability) | Acres | Percent of Area |
|---|---|---|
| Not predicted to experience wildfire (0% probability) | 14,170,411 | 8.2% |
| Low (>0 - 0.01% probability) | 6,905,711 | 4.0% |
| Low-moderate (>0.01 - 0.02154% probability) | 3,358,398 | 2.0% |
| Low-moderate (>0.02154 - 0.04642% probability) | 6,863,262 | 4.0% |
| Moderate (>0.04642 - 0.1% probability) | 24,328,517 | 14.2% |
| Moderate (>0.1 - 0.21544% probability) | 43,057,578 | 25.0% |
| Moderate (>0.21544 - 0.46416% probability) | 31,454,604 | 18.3% |
| Moderate-high (>0.46416 - 1% probability) | 23,023,414 | 13.4% |
| Moderate-high (>1 - 2.15443% probability) | 14,529,697 | 8.5% |
| High (>2.15443 - 4.64159% probability) | 4,054,247 | 2.4% |
| High (>4.64159% probability) | 129,016 | <0.1% |
| *No wildfire risk data available* | *22,569* | *<0.1%* |
| **Total area** | **171,897,425** | **100%** |